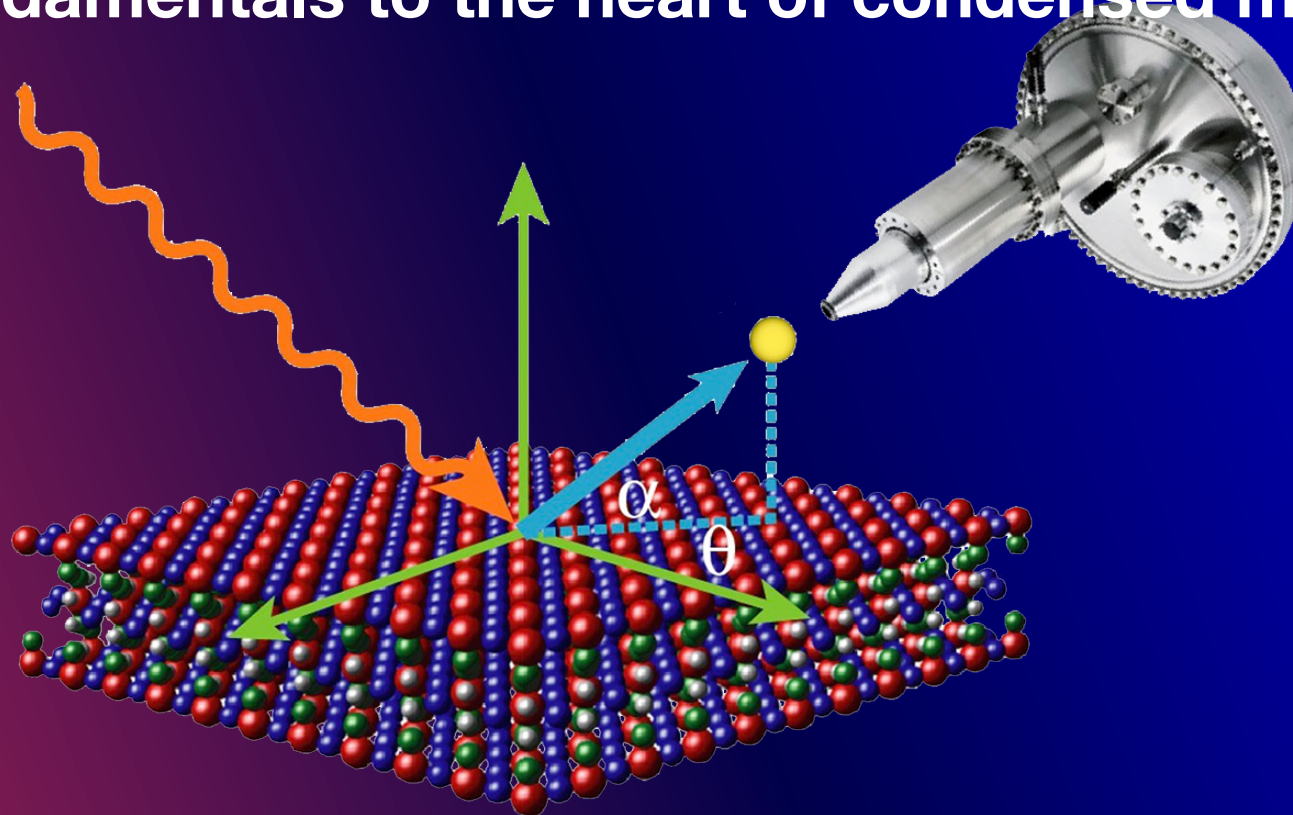


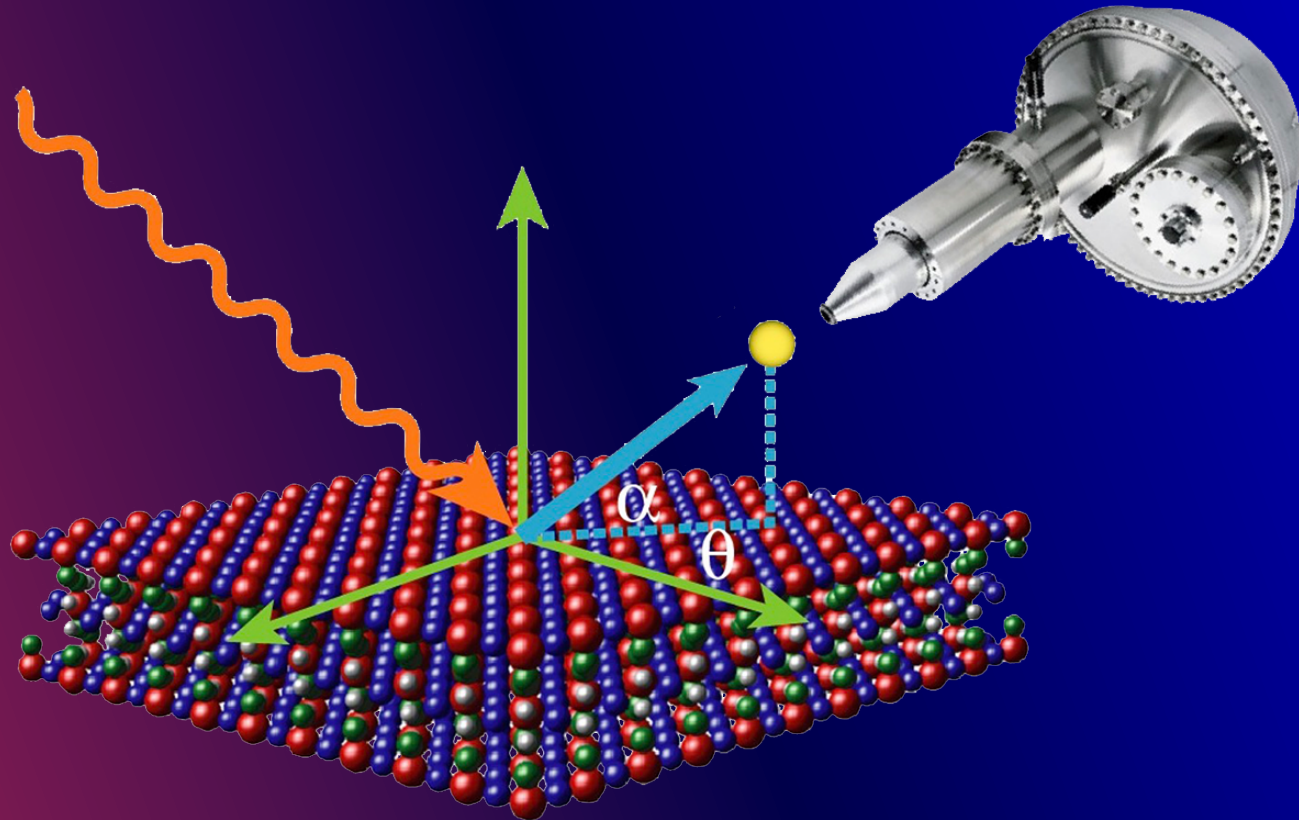
Angle Resolved Photoemission Spectroscopy

From fundamentals to the heart of condensed matter



6-7 FEBRERO, 2023

Lecture #1 : From Einstein to Band Mapping



6-7 FEBRERO, 2023

Overview of Lectures & Labs

Monday, February 6

- 9:30 - 11:00 Lecture #1 : From Einstein to Band Mapping
- 11:30 -13:00 Lecture #2 : From Particles to Quasiparticles

Tuesday, February 7

- 9:00 – 12:30 Visit to ALBA Synchrotron & ARPES demonstration at LOREA beamline
- 14:30 - 16:00 Lecture #3 : ARPES Studies of Quantum Materials
- 16:30 -18:00 Lecture #4 : Frontiers in ARPES – Time, Space, and Spin

Organizers

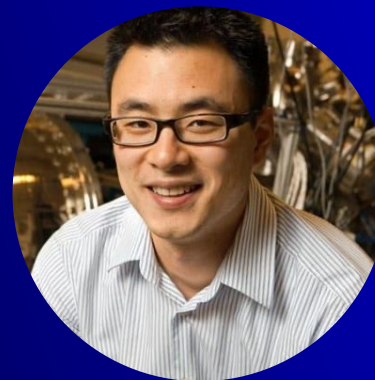


Gervasi Herranz
MULFOX, ICMAB



Josep Fontcuberta
MULFOX, ICMAB

Presenters



Kyle Shen
Cornell Univ. & ICMAB

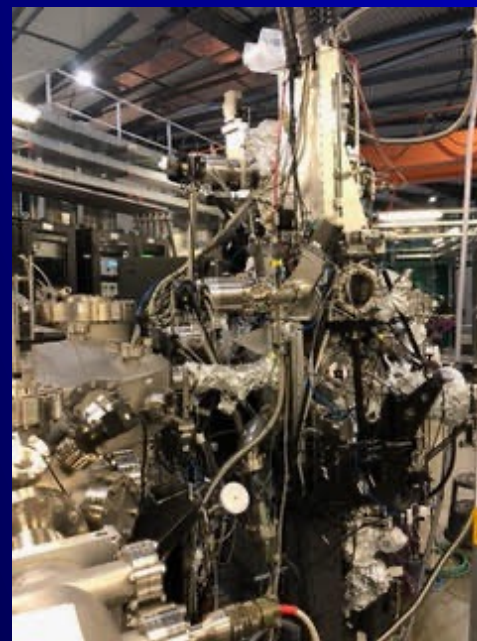


Massimo Tallarida
ALBA Synchrotron

A few important points!

- Participation is *highly encouraged & important!*
Please feel free to ask questions *any time!*
- A goal of our school is to encourage more local users
at LOREA. *This means you!*
- Remember to bring your identification (DNI / TIE /
Passport) tomorrow if you are registered for the ALBA
visit
- If you have further questions, I am very happy to meet around
ICMAB / UAB! My email is : kmshen@cornell.edu

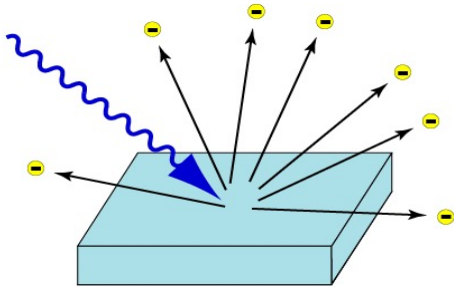
LOREA Beamline @ ALBA



- High spatial resolution (~ 20 microns spot size with microscope)
- Advanced sample preparation capabilities (annealing, vacuum transfer)
 - Spin-resolved detection (installed & coming online this spring)
 - Wide photon energy range (20 – 500+ eV)

Your scientific background is best described as :

- A. Materials Science & Engineering
- B. Condensed Matter Physics
- C. Electrical Engineering
- D. Solid State Chemistry
- E. Other



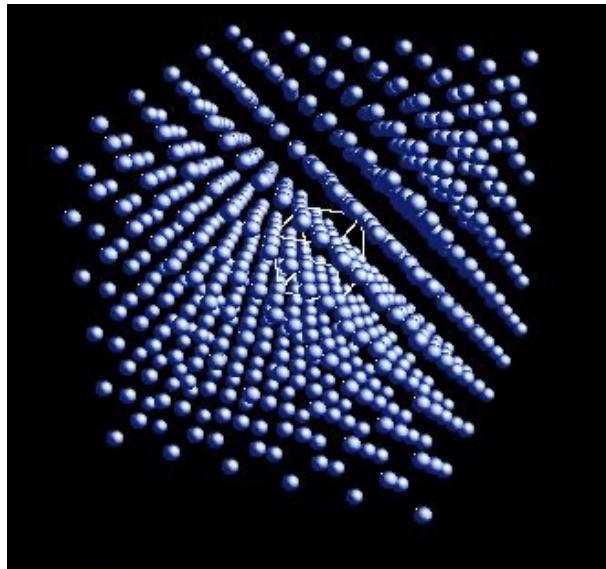
Angle Resolved PhotoElectron Spectroscopy

FIRST EVIDENCE FOR THE QUANTIZATION OF LIGHT!

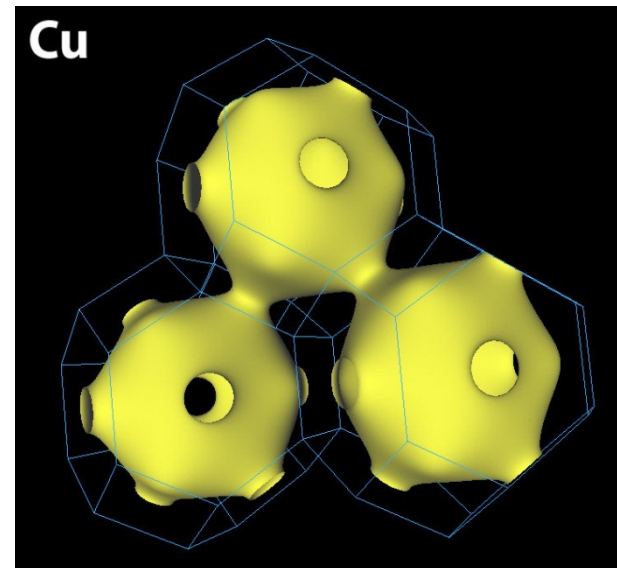
Velocity and direction of the electrons in the solid

Low-energy Electronic Structure → Macroscopic Physical Properties

Superconductivity, Magnetism, Density Waves,



X-ray diffraction

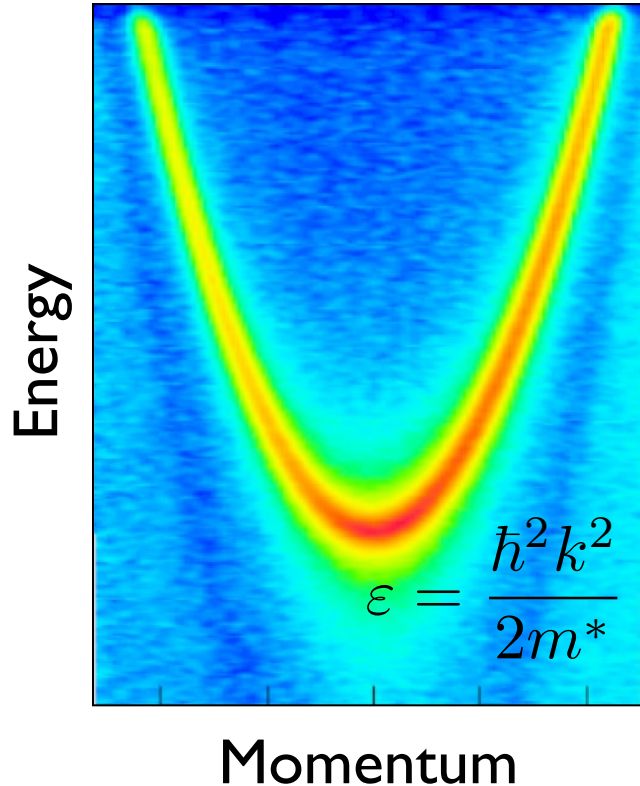


Photoemission

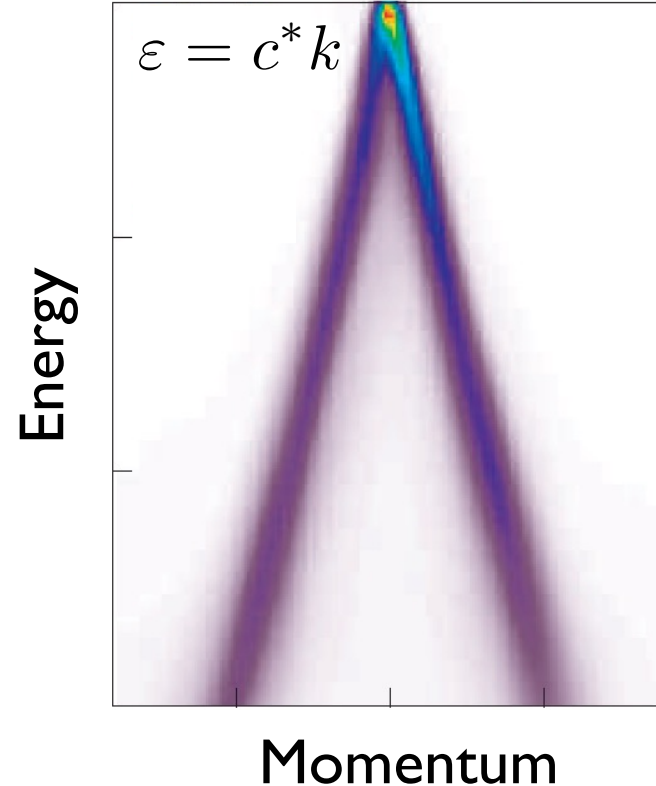
Measuring quasiparticle band velocities & band masses

$$\hbar v_k = \frac{\partial \varepsilon}{\partial k}$$

$$m_k^{-1} = \left(\hbar^{-2} \frac{\partial^2 \varepsilon}{\partial k^2} \right)$$

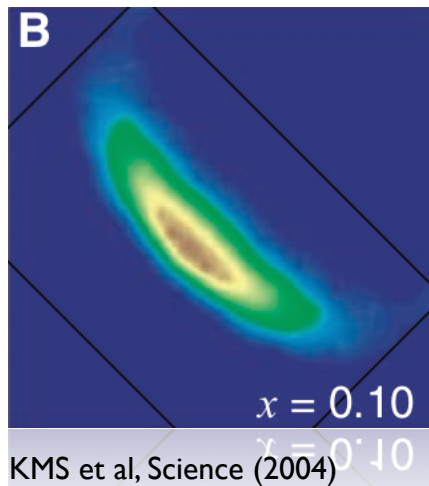


Reinert & Hufner NJP 2005

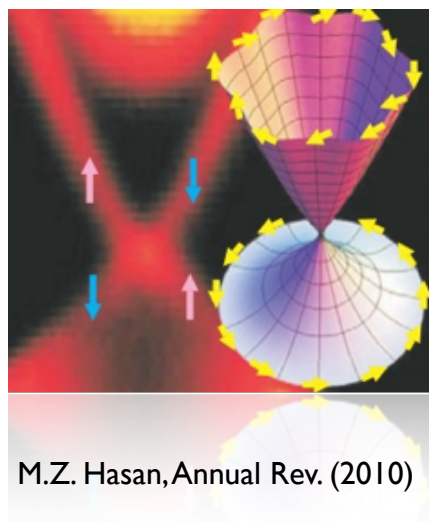


Zhou et al., Nature Physics 2006

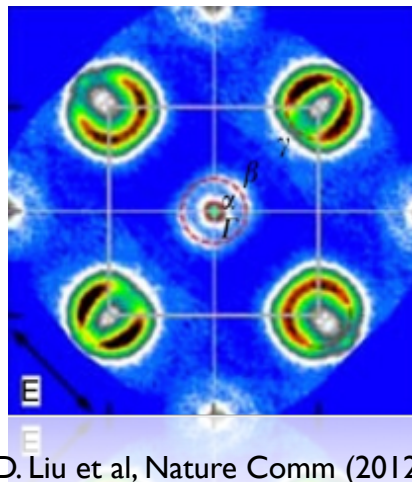
Cuprate Superconductors



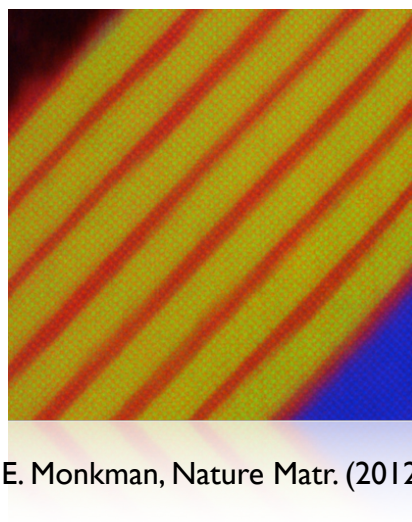
Topological Materials



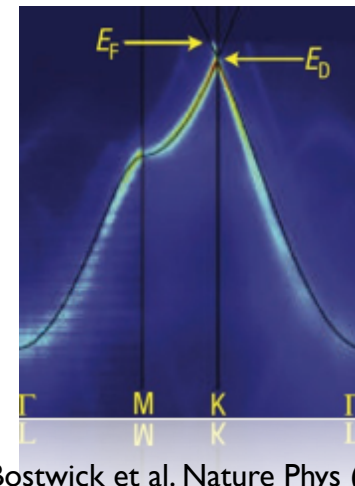
Fe-based Superconductors



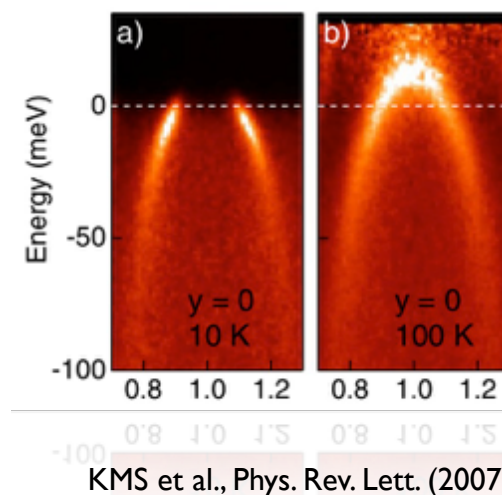
Superlattices & Interfaces



2D Materials



Complex Oxides





- The Brownian motion

"On the motion of small particles suspended in liquids at rest required by the molecular-kinetic theory of heat."
Annalen der Physik, 17 (1905), pp. 549-560.

- **The photoelectric effect**

"On a heuristic viewpoint concerning the production and transformation of light"
Annalen der Physik, 17 (1905), pp. 132-148.



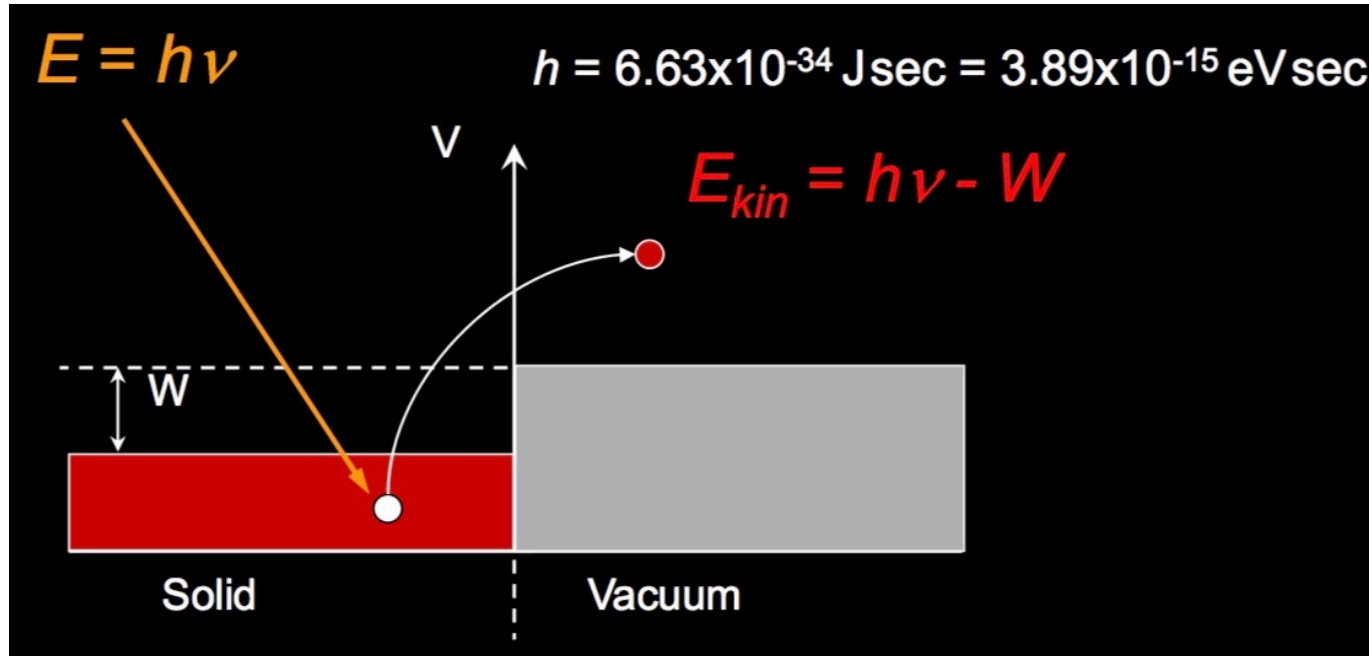
- The special theory of relativity

"On the electrodynamics of moving bodies"
Annalen der Physik, 17 (1905), pp. 891-921

- Mass-energy Equivalency $E=mc^2$

"Does the inertia of a body depend on its energy?"
Annalen der Physik, 18 (1905), pp. 639-41.

1905 : Einstein' hypothesis: light quanta with $E = h\nu = hc / \lambda$



The maximum $E_{kin} = \frac{1}{2}mv^2$ is proportional to the FREQUENCY but depends also on the material work function W

The NUMBER of electrons is proportional only to the INTENSITY

In 1913 Einstein was elected to the Prussian Academy of Sciences and appointed to a research position in Berlin. In his nomination speech to the Prussian Academy, Planck says:

*"Summing up, we may say that there is hardly one among the great problems in which modern physics is so rich, to which Einstein has not made an important contribution. **That he may sometimes have missed the target in his speculations, as for example, in his hypothesis of light quanta, cannot really be held too much against him**, for it is not possible to introduce fundamentally new ideas, even in the most exact sciences, without occasionally taking a risk".*

1887 : discovery of photoelectric effect by Hertz

1905 : quantization of light by Einstein : $E_{\max} = h\nu - \phi$



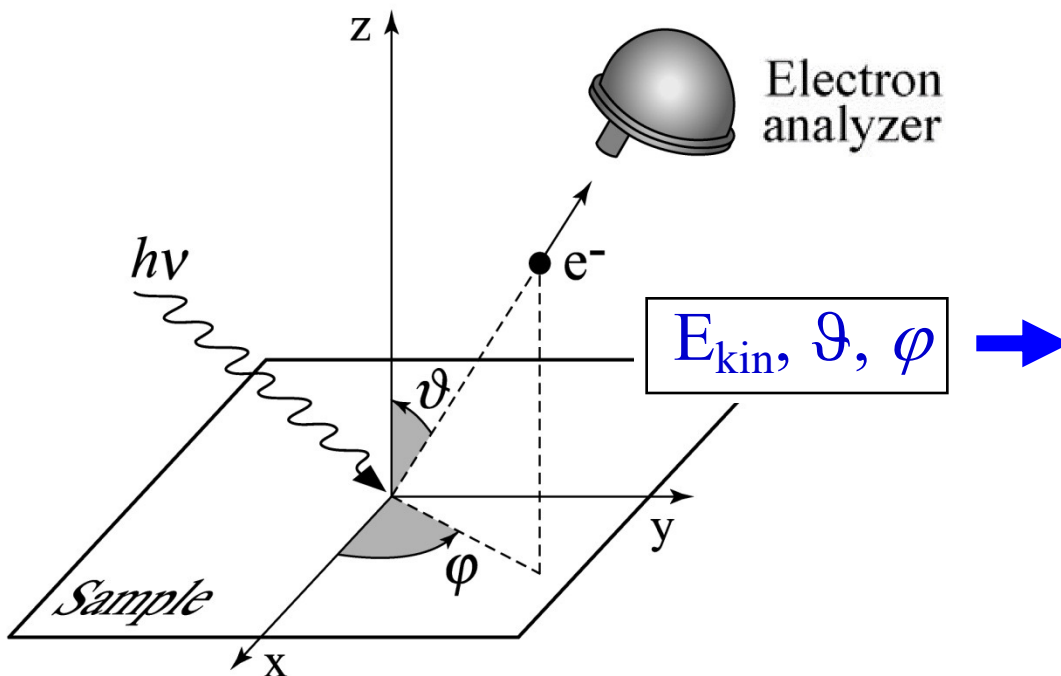
1957 : x-ray photoelectron spectroscopy for chemical analysis (XPS or ESCA) by Siegbahn and others



1960s : ultraviolet photoemission spectroscopy (UPS) by Spicer, Turner, others

1970s : angle-resolved photoemission spectroscopy (ARPES) by Spicer, Smith, others

1980s – present day : widely used as materials characterization tool



$$\mathbf{K} = \mathbf{p} / \hbar = \sqrt{2mE_{kin}} / \hbar$$

$$K_x = \frac{1}{\hbar} \sqrt{2mE_{kin}} \sin \vartheta \cos \varphi$$

$$K_y = \frac{1}{\hbar} \sqrt{2mE_{kin}} \sin \vartheta \sin \varphi$$

$$K_z = \frac{1}{\hbar} \sqrt{2mE_{kin}} \cos \vartheta$$

Vacuum

$$\begin{matrix} E_{kin} \\ \mathbf{K} \end{matrix}$$

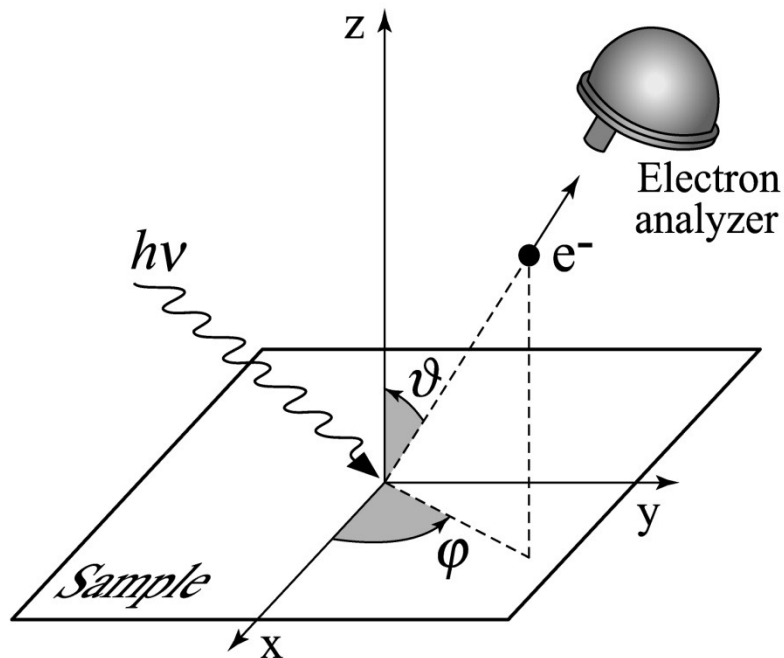
Conservation laws

$$\begin{aligned} E_f - E_i &= h\nu \\ \mathbf{k}_f - \mathbf{k}_i &= \mathbf{k}_{h\nu} \end{aligned}$$

Solid

$$\begin{matrix} E_B \\ \mathbf{k} \end{matrix}$$

Kinematics of the ARPES process : Energy Conservation **ARPES2023**

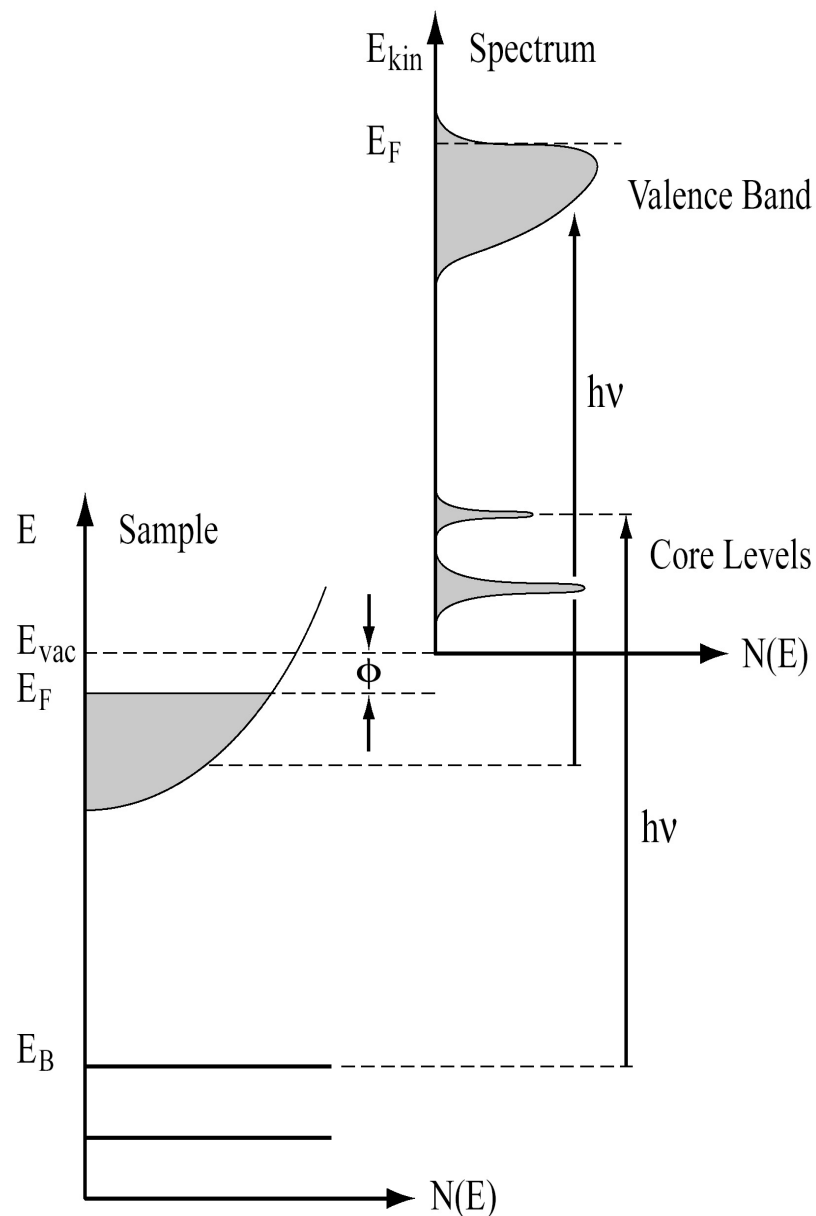


Energy Conservation

$$E_{kin} = h\nu - \phi - |E_B|$$

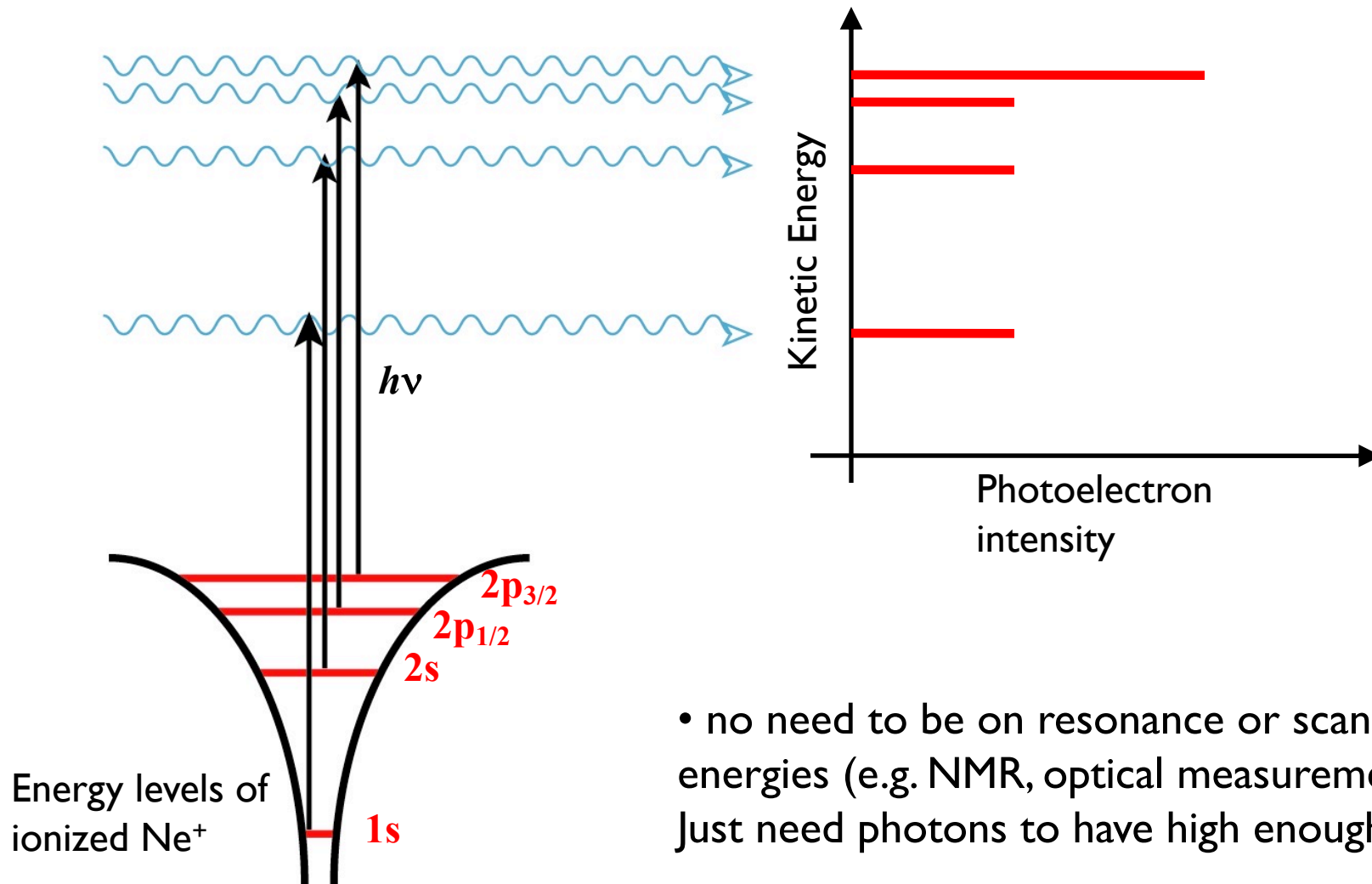
Momentum Conservation

$$\hbar \mathbf{k}_{||} = \hbar \mathbf{K}_{||} = \sqrt{2m E_{kin}} \cdot \sin \theta$$



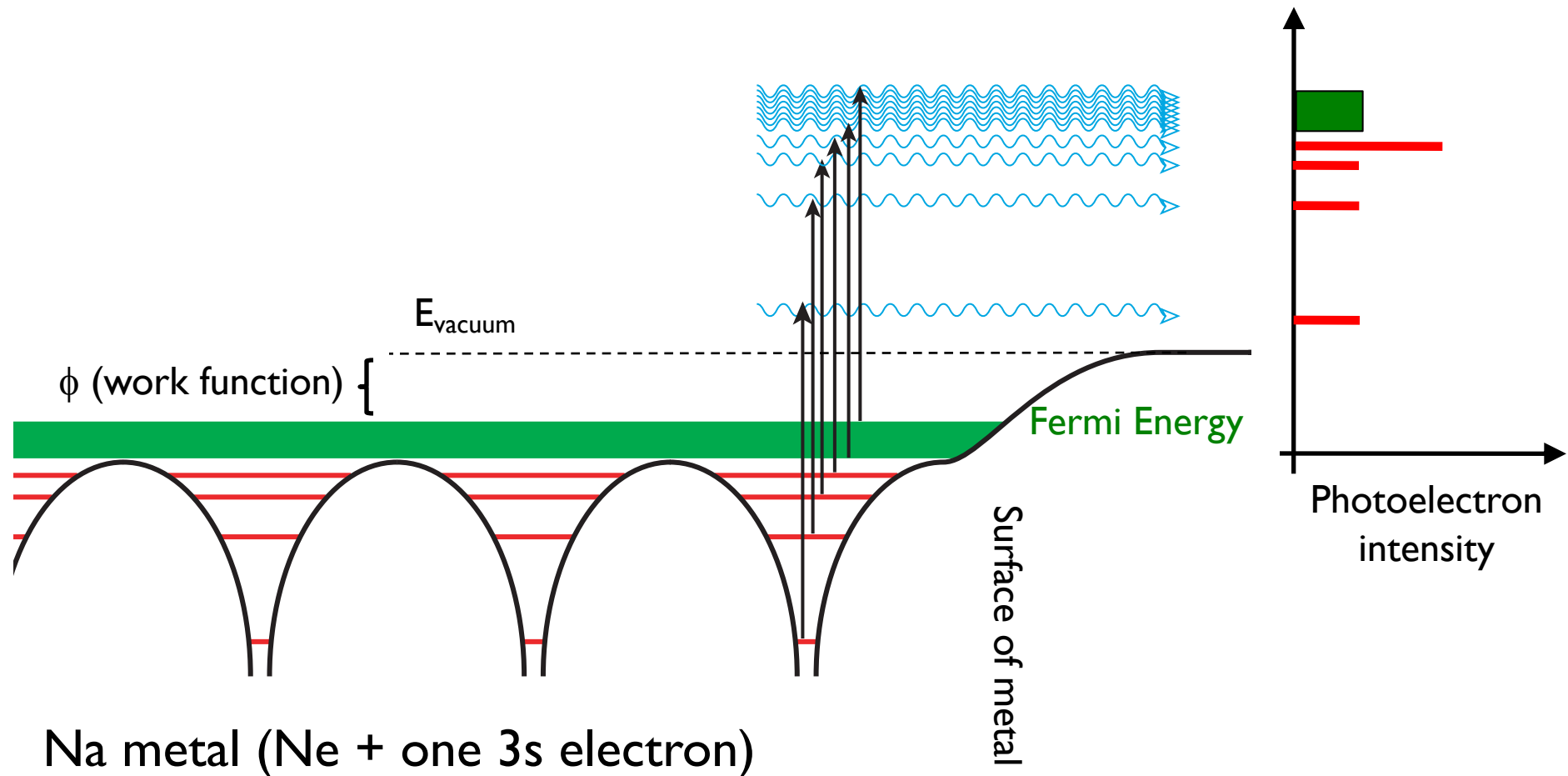
Electron ionization levels are formed from discrete levels in an atom

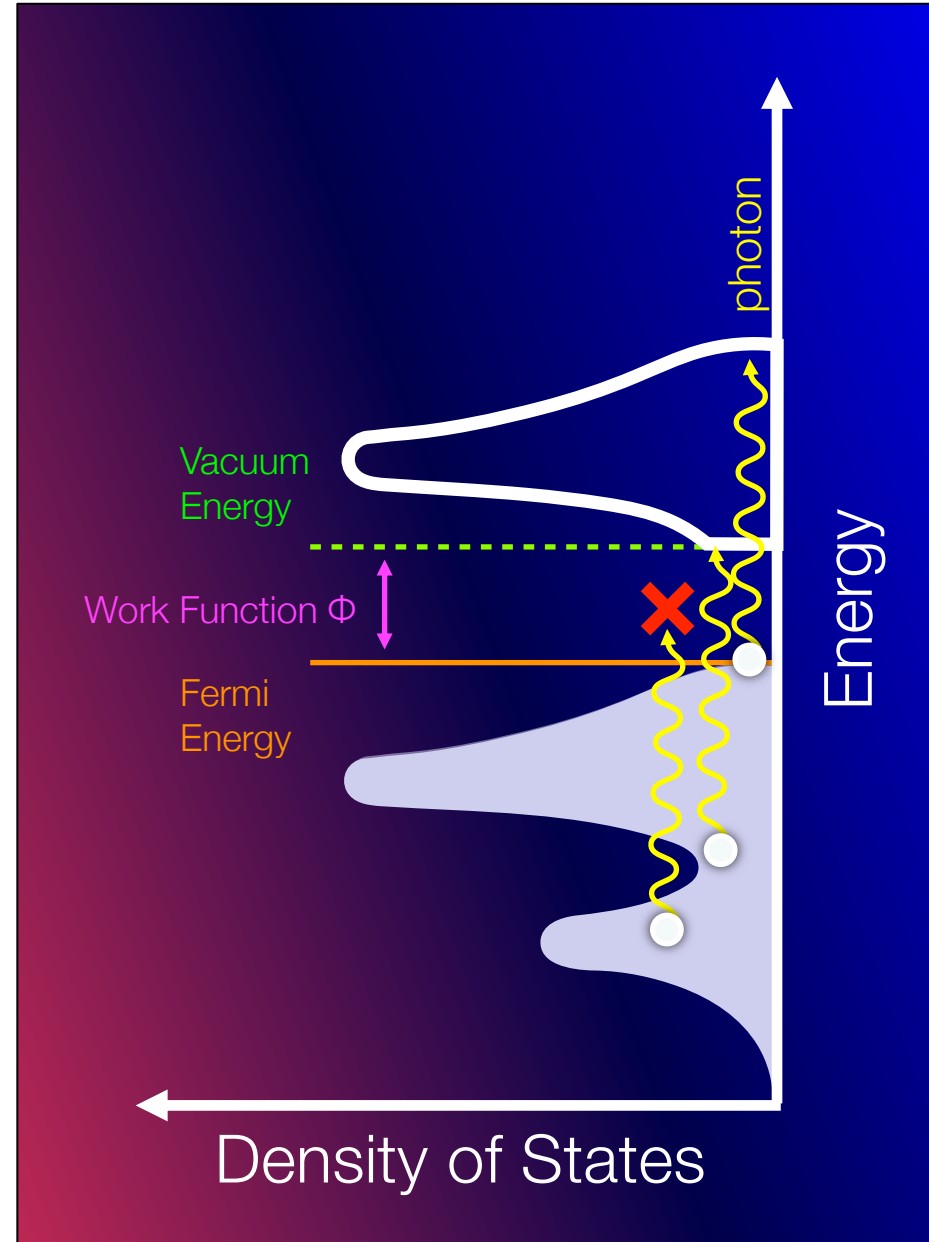
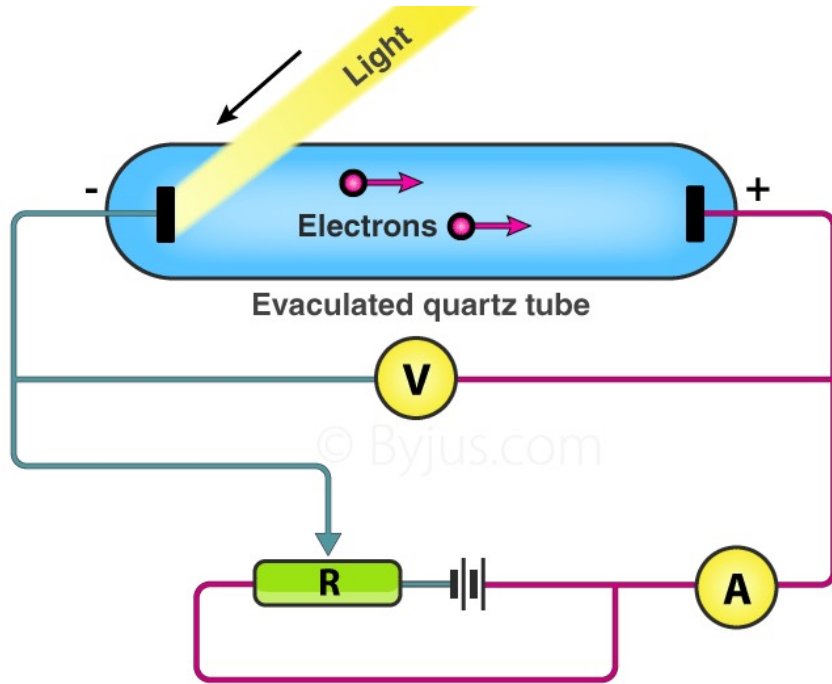
Energy conservation: $h\nu - E_I = KE$



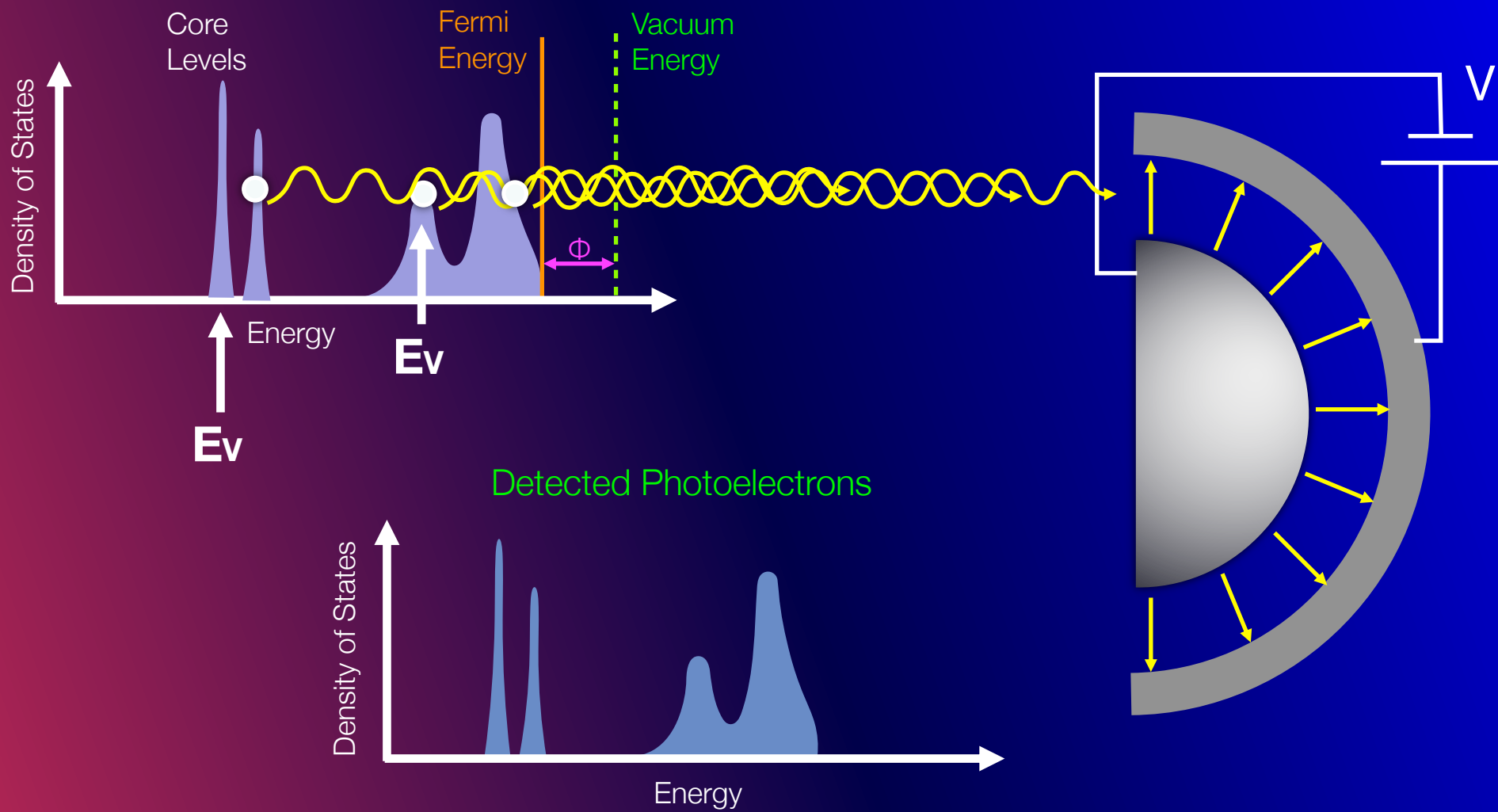
- no need to be on resonance or scan photon energies (e.g. NMR, optical measurements). Just need photons to have high enough energy!

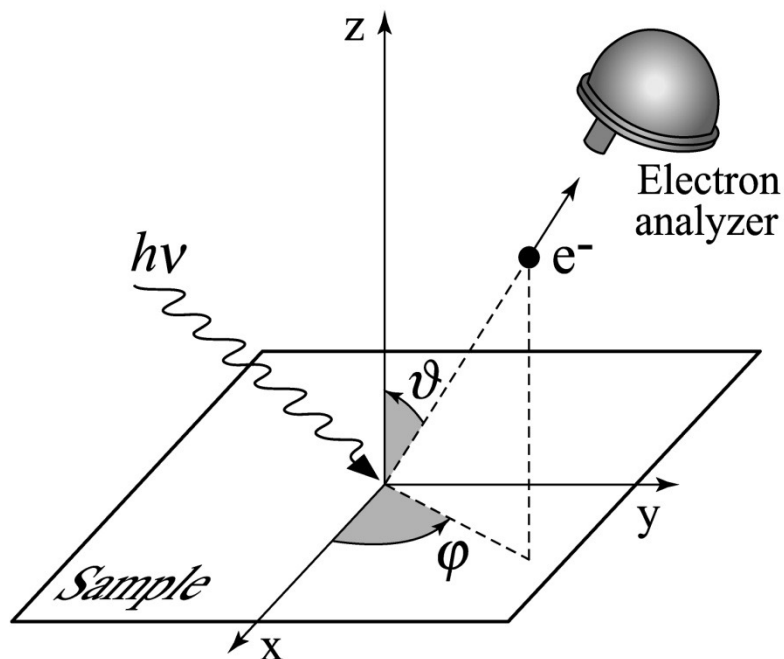
- Deeply bound “*core*” electrons remain basically unchanged
- Outermost “*valence*” electrons hybridize forming continuous “*energy bands*”



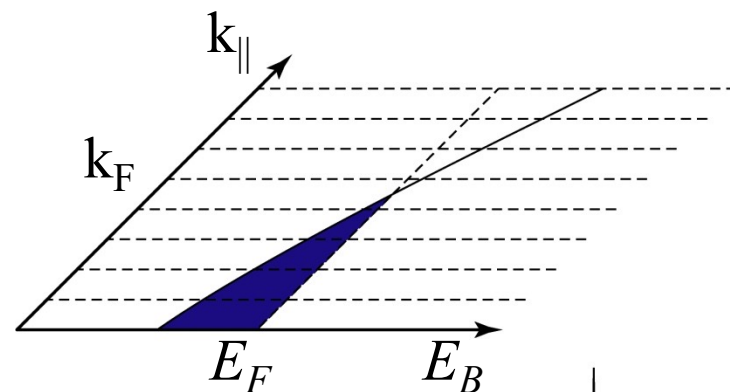


Energy analysis of photoelectrons gives density of states **ARPES**2023





Electrons in Reciprocal Space

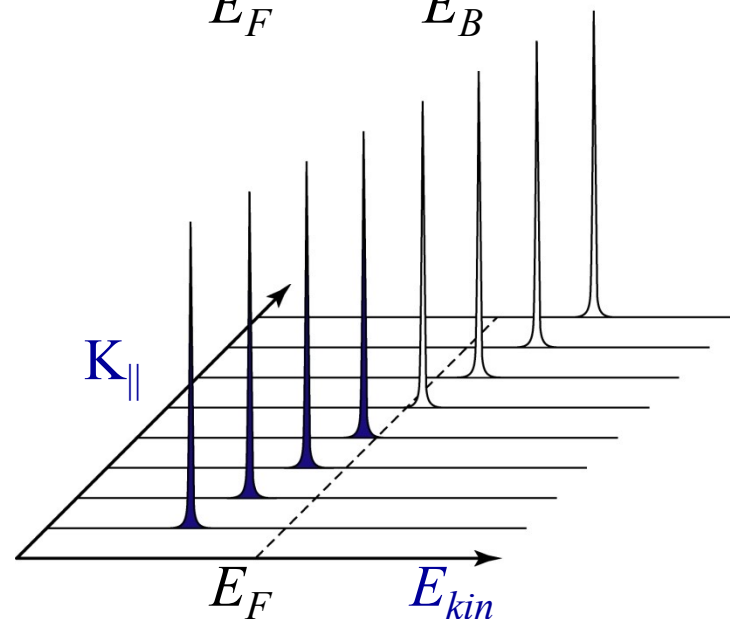


Energy Conservation

$$E_{kin} = h\nu - \phi - |E_B|$$

Momentum Conservation

$$\hbar k_{||} = \hbar K_{||} = \sqrt{2m E_{kin}} \cdot \sin \theta$$



towards a quantum theory of solids

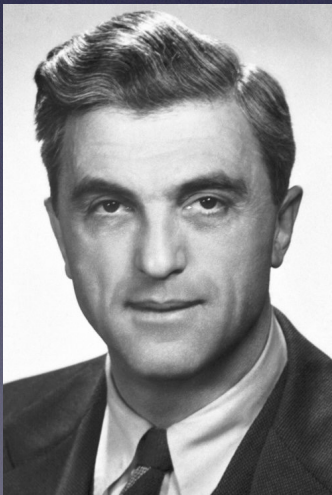


Erwin
Schrodinger

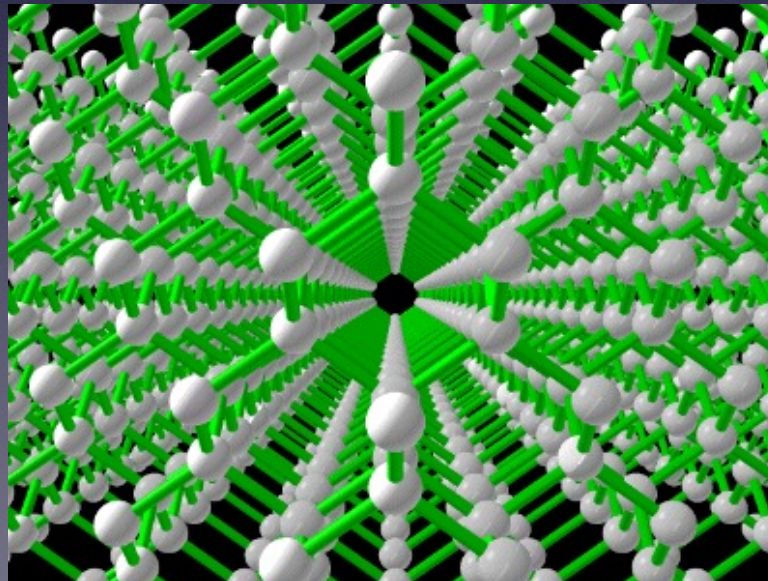
Schrodinger Equation

$$\frac{-\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + U(x)\psi(x) = E\psi(x)$$

time-independent in 1D



Felix Bloch



“When I started to think about it, I felt that the main problem was to explain how the electrons could sneak by all the ions in a metal ... By straight Fourier analysis I found to my delight that the wave differed from the plane wave of free electrons only by a periodic modulation”

- Felix Bloch

towards a quantum theory of solids



Erwin
Schrodinger

Schrodinger Equation

$$\frac{-\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + U(x)\psi(x) = E\psi(x)$$

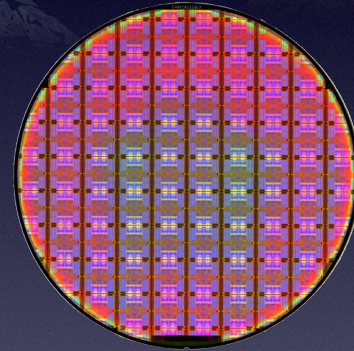
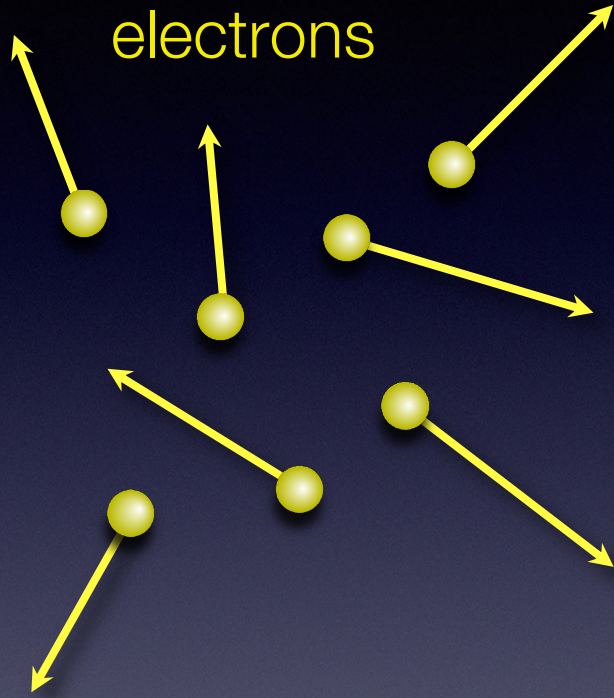
time-independent in 1D

the Hamiltonian for interacting electrons & nuclei is
basically unsolvable!

electron KE	electron-electron interactions	nuclei KE	electron-nuclei interactions	nuclei-nuclei interactions
$-\sum_j^{N_e} \frac{\hbar^2}{2m} \nabla_j^2$	$+\sum_{j \ll k}^{N_e} \frac{e^2}{ \vec{r}_j - \vec{r}_k }$	$-\sum_{\alpha}^{N_i} \frac{\hbar^2}{2M_{\alpha}} \nabla_{\alpha}^2$	$-\sum_j^{N_e} \sum_{\alpha}^{N_i} \frac{Z_{\alpha} e^2}{ \vec{r}_j - \vec{R}_{\alpha} }$	$+\sum_{\alpha \ll \beta}^{N_j} \frac{Z_{\alpha} Z_{\beta} e^2}{ \vec{R}_{\alpha} - \vec{R}_{\beta} }$

the “independent electron” approximation

independent
electrons



electron
KE

electron-electron
interactions

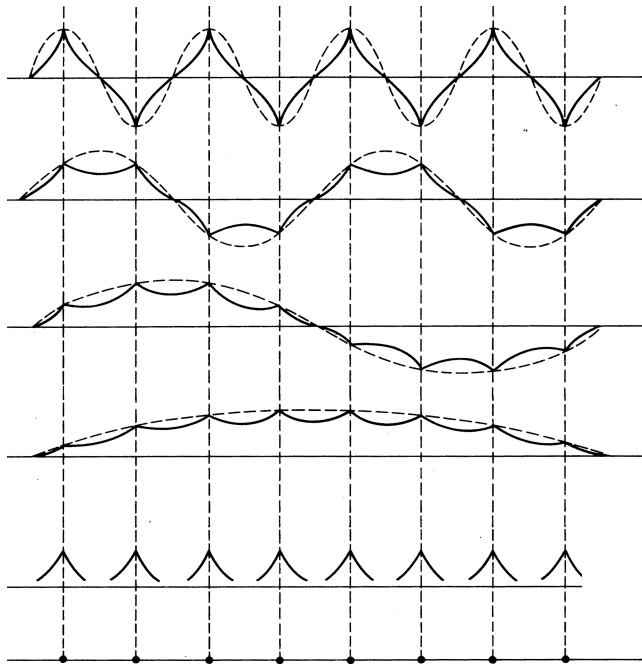
nuclei
KE

electron-nuclei
interactions

nuclei-nuclei
interactions

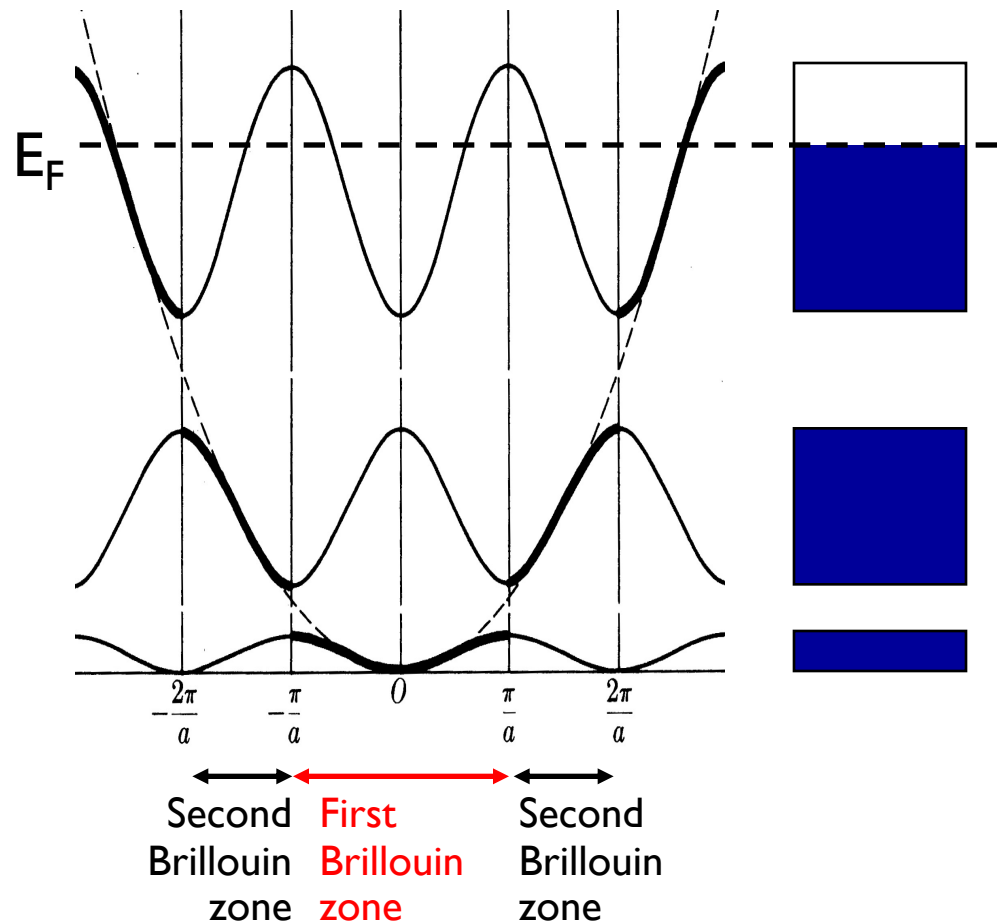
$$\begin{aligned}
 & - \sum_j^{N_e} \frac{\hbar^2}{2m} \nabla_j^2 + \sum_{j \ll k}^{N_e} \frac{e^2}{|\vec{r}_j - \vec{r}_k|} - \sum_{\alpha}^{N_i} \frac{\hbar^2}{2M_{\alpha}} \nabla_{\alpha}^2 - \sum_j^{N_e} \sum_{\alpha}^{N_i} \frac{Z_{\alpha} e^2}{|\vec{r}_j - \vec{R}_{\alpha}|} + \sum_{\alpha \ll \beta}^{N_j} \frac{Z_{\alpha} Z_{\beta} e^2}{|\vec{R}_{\alpha} - \vec{R}_{\beta}|}
 \end{aligned}$$

Wave functions in a 1D lattice

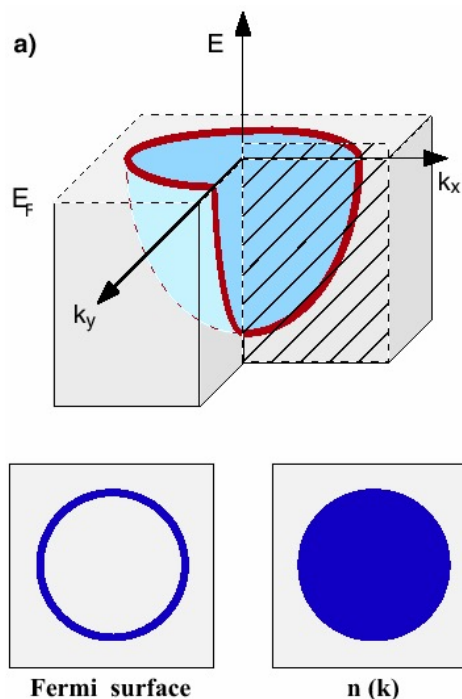


1D chain of atoms

Allowed electronic states Repeated-zone scheme

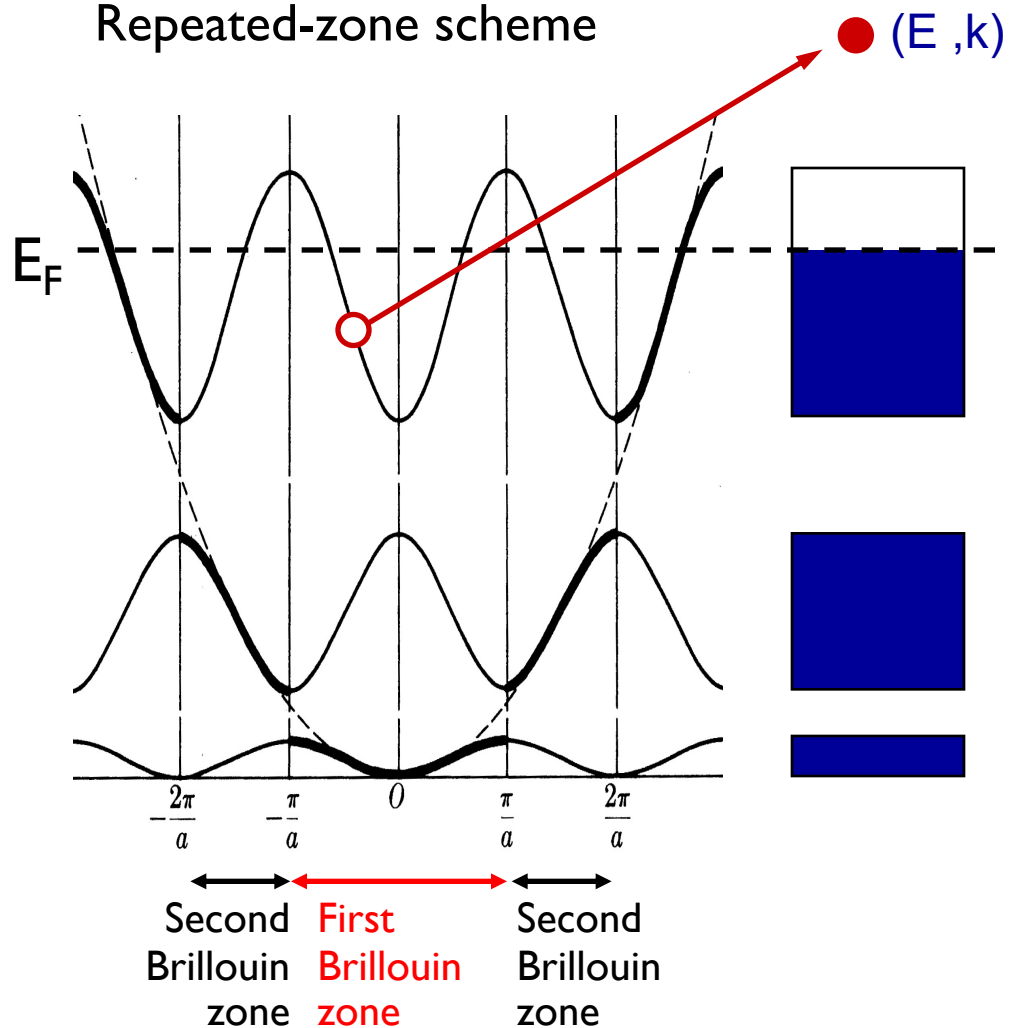


Many **properties** of a solids are determined by **electrons near E_F** (conductivity, magnetoresistance, superconductivity, magnetism)

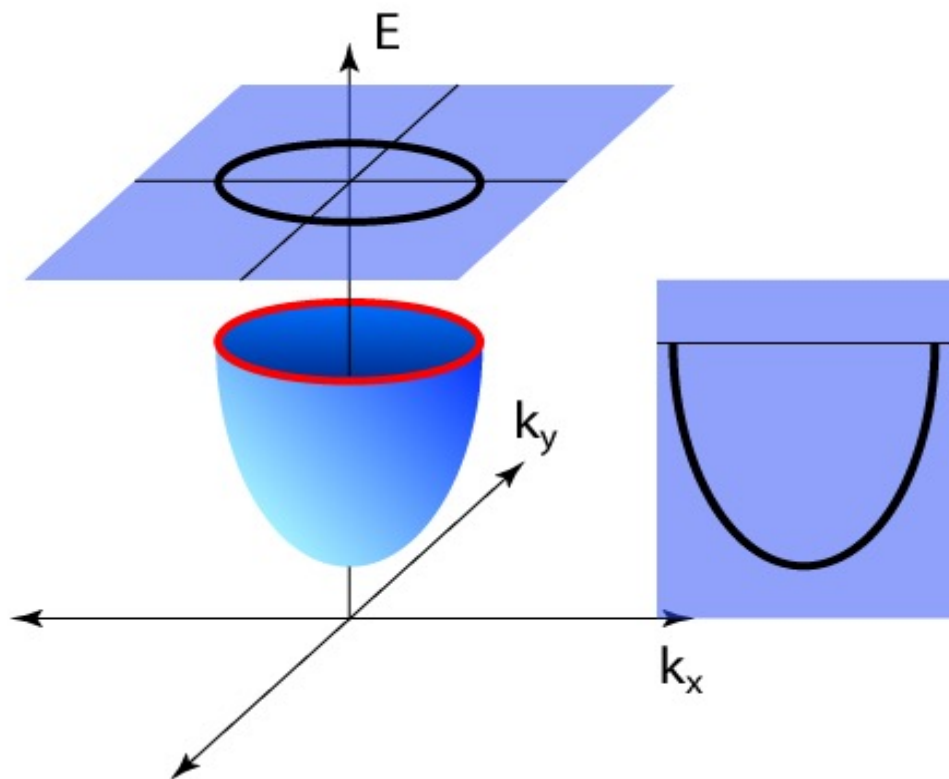


Only a **narrow energy slice** around E_F is relevant for these properties ($kT=25$ meV at room temperature)

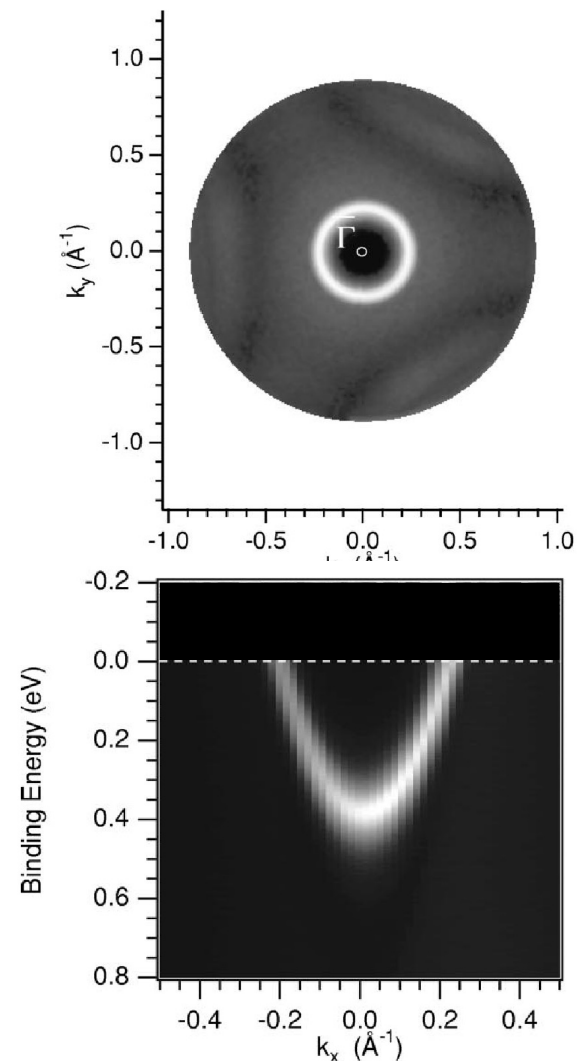
Allowed electronic states
Repeated-zone scheme



Cu (111)



$$E = \hbar^2 \mathbf{k}^2 / 2m^*$$



F. Baumberger et al., PRB 64, 195411 (2001)

In the photoemission process, energy conservation allows us to relate the energy of the exiting photoelectron to the initial state energy of the electron by considering the energy of the incident photon.

When considering momentum conservation in the photoemission process, which statement is **true**?

- A. In a photon-in, electron-out experiment, the momentum of the incident photon never needs to be considered (in contrast, to photon-in, photon-out, like x-ray scattering)
- B. The photon's momentum does not need to be considered because translational symmetry is broken in the perpendicular direction
- C. Due to its relativistic nature, the photon's momentum is typically negligibly small when compared to the electron's momentum, and can therefore be ignored in an ARPES experiment
- D. The photon's momentum affects the ARPES spectra in the form of momentum broadening, Δk
- E. None of the above statements are true

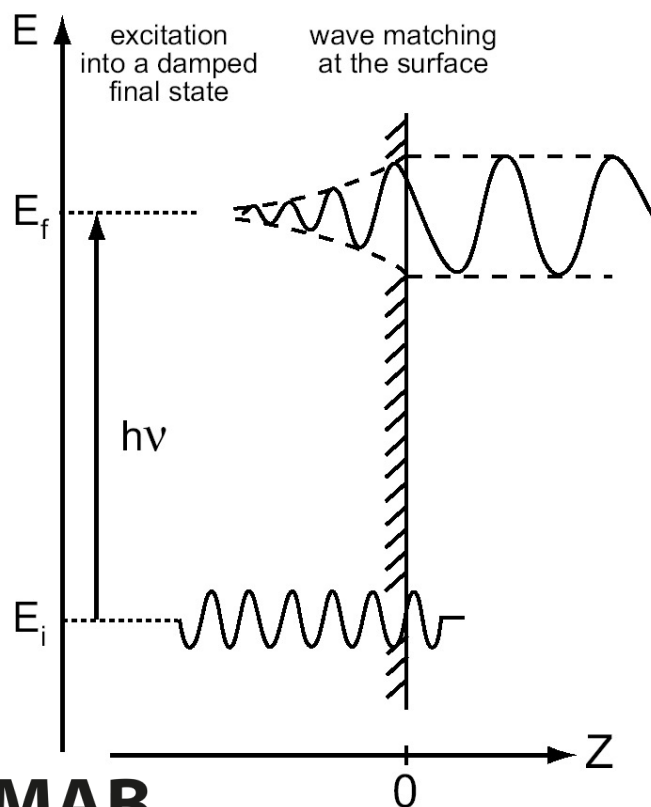
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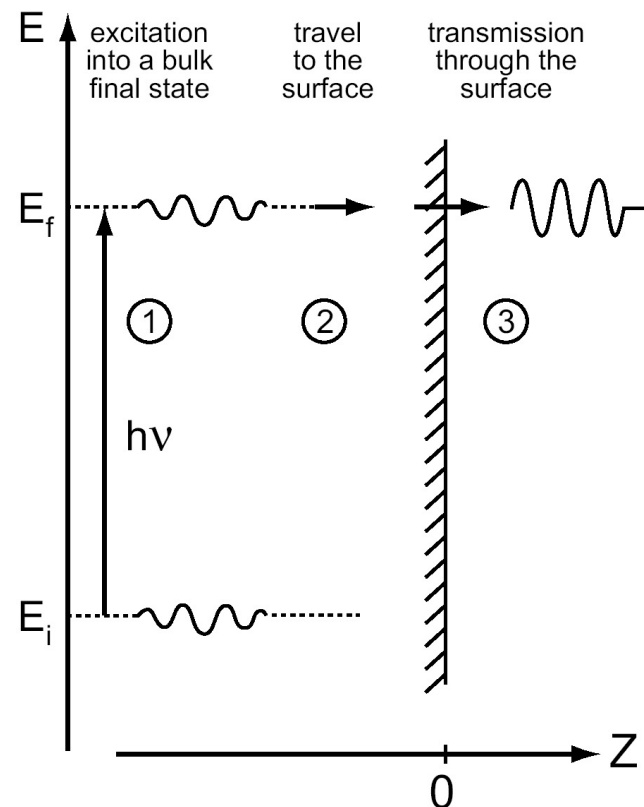
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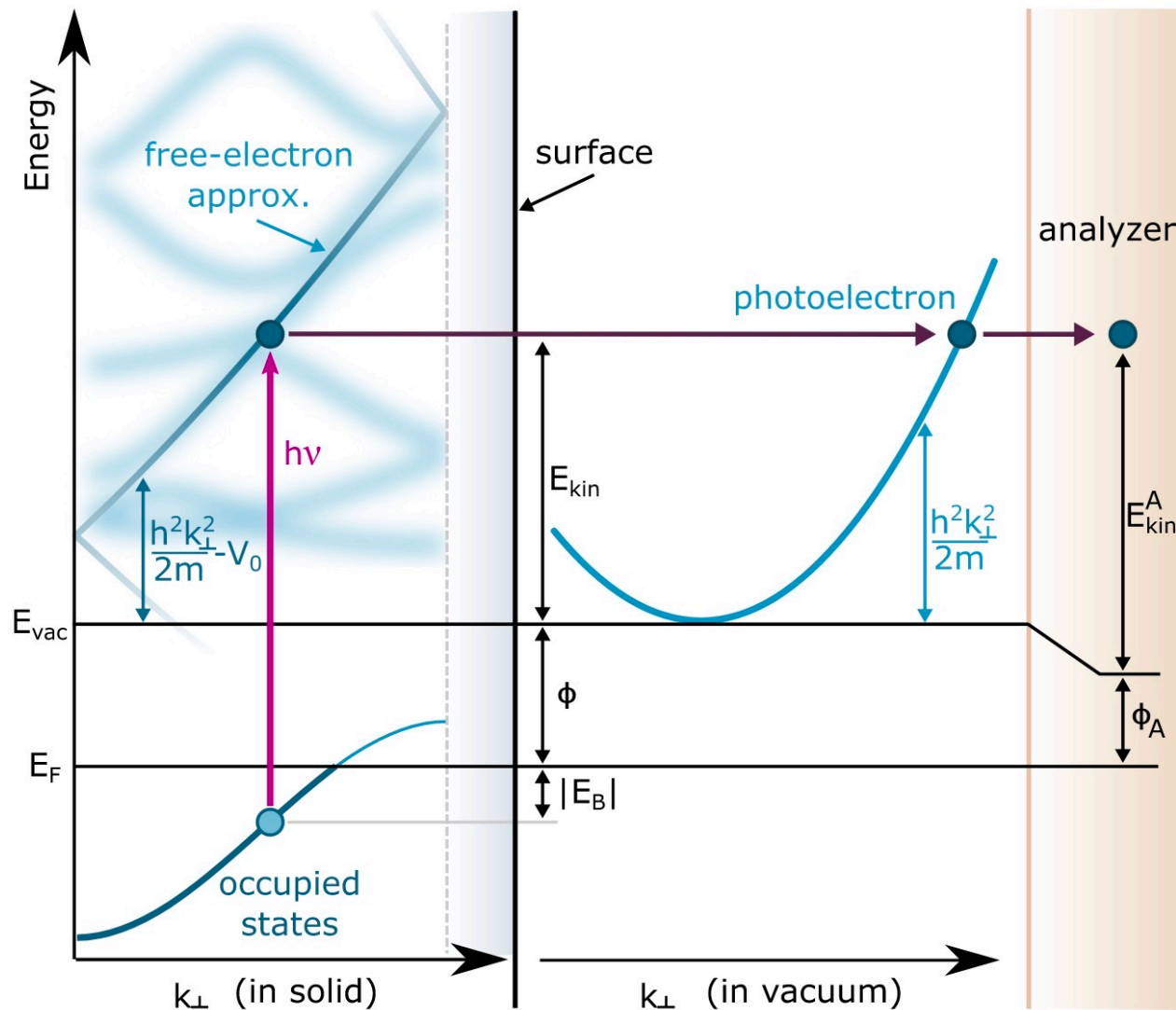
Photoemission Intensity $I(k, \omega)$ $\left\{ \begin{array}{l} w_{fi} \propto |\langle \Psi_f^N | \mathbf{A} \cdot \mathbf{p} | \Psi_i^N \rangle|^2 \delta(E_f^N - E_i^N - h\nu) \end{array} \right.$

One-step model

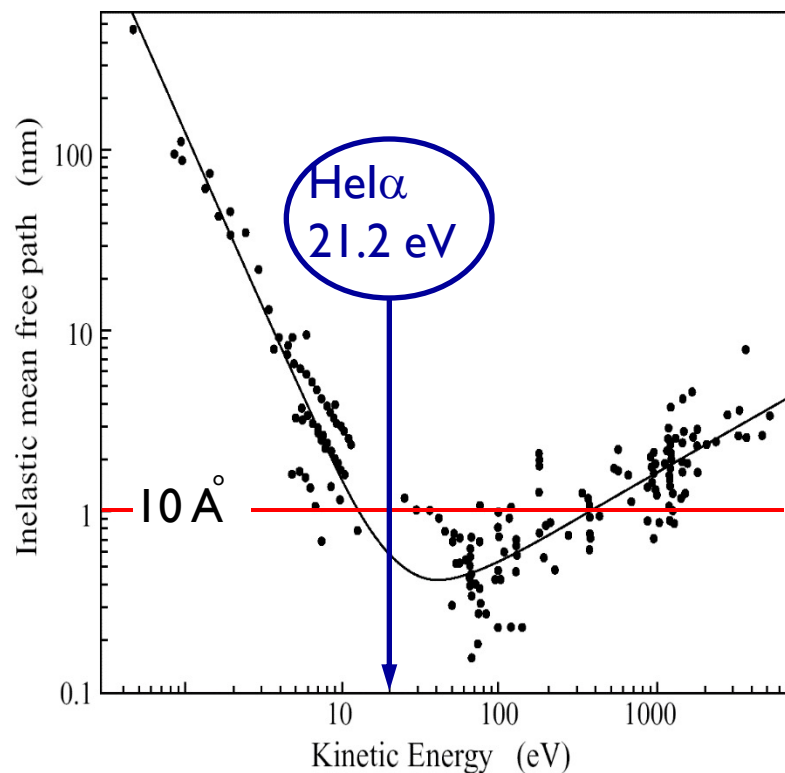


Three-step model

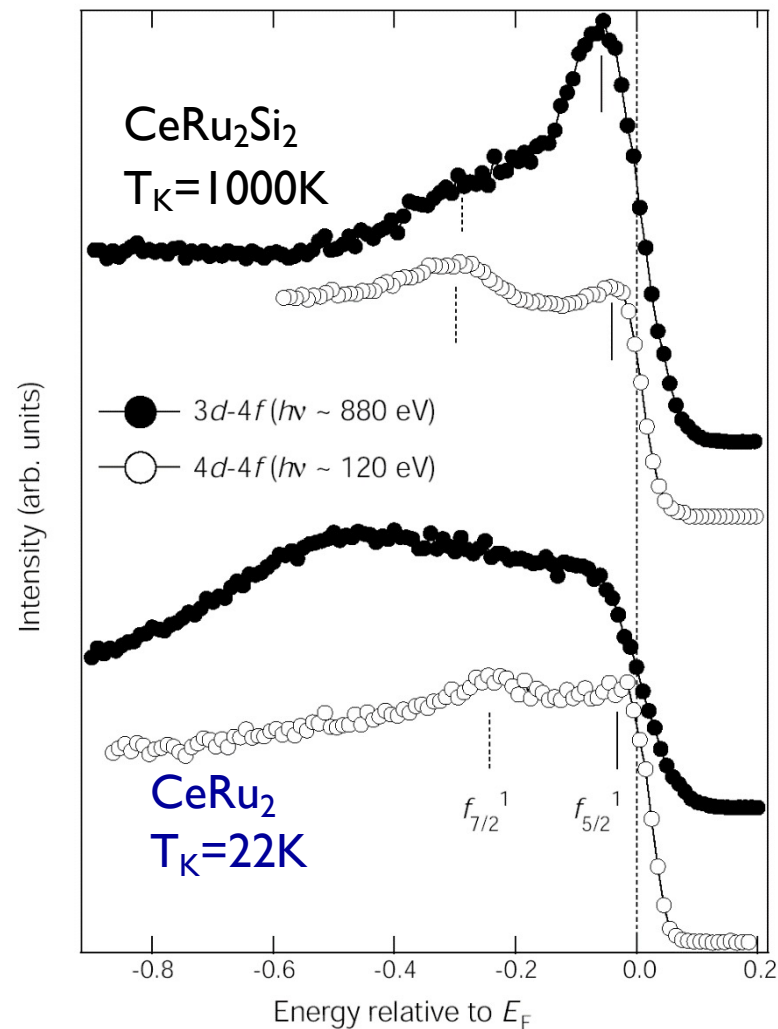




“Universal Curve” of Mean-free path for excited electrons



Seah, Dench *et al.*, SIA 1, 2 (1979)



Sekiyama *et al.*, Nature 403, 396 (2000)

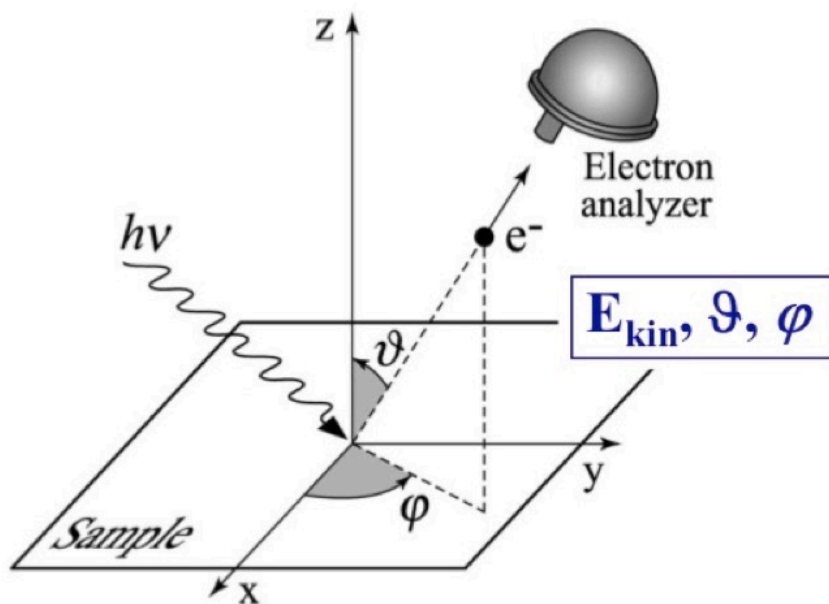
Photoemission is a highly surface sensitive measurement (typically < 1 nm probing depth). What are the main factor(s) determining the probing depth (inelastic mean free path) in the photoemission process?

- | | |
|---|---------------------|
| 1. Adsorbed molecules on the surface of the sample | A. 1, 3, 5 & 6 |
| 2. The absorption length (skin depth) of photons in the VUV | B. 3 & 4 |
| 3. The kinetic energy of the photoelectrons | C. 2, 3, 4 & 5 |
| 4. The plasma frequency of the measured sample | D. 2, 3 & 5 |
| 5. The polarization of the photons being used to measure the sample | E. All of the above |

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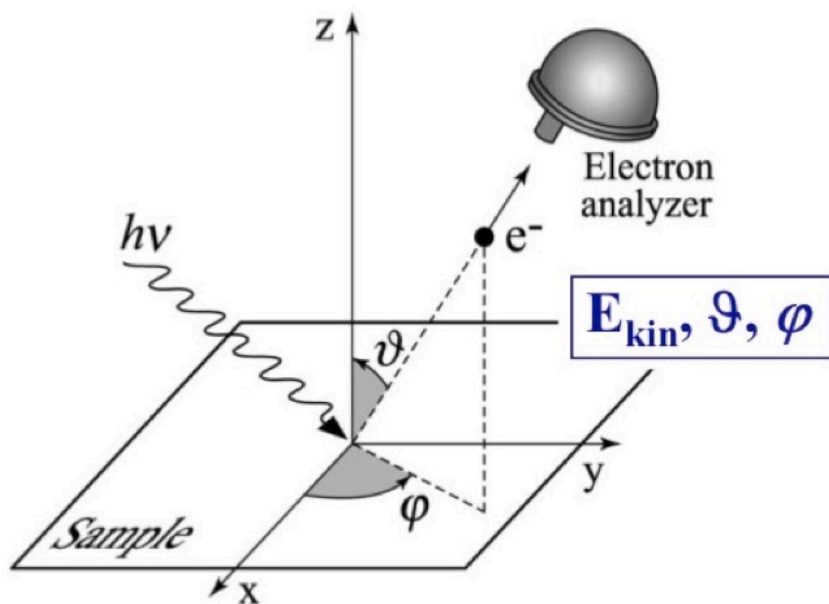
- | | |
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In the photoemission process, the in-plane (longitudinal) momentum of the photoelectron can be directly related to the in-plane momentum of the electron when it was inside the solid, due to translational symmetry. For the out-of-plane (perpendicular) component of the photoelectron's momentum :



- A. The out-of-plane momentum of the electron in the solid is equal to the out-of-plane momentum of the outgoing photoelectron
- B. The perpendicular component of the electron's momentum cannot be experimentally determined
- C. The perpendicular momentum of the electron inside the crystal is not a good quantum number
- D. The out-of-plane photoelectron momentum can be related to the out-of-plane momentum of the electron when it was in the solid, but they are not precisely equal

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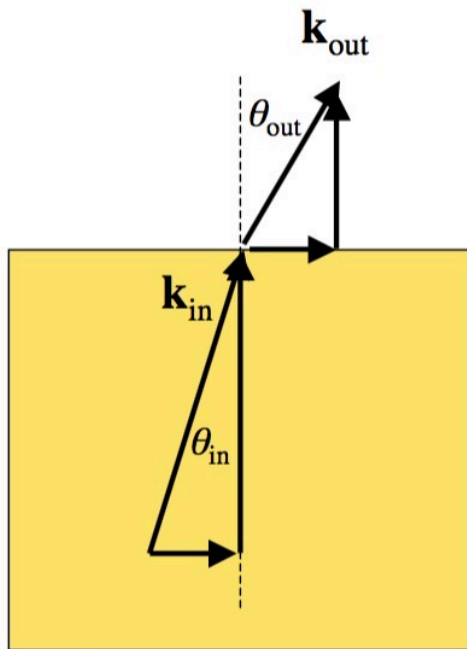


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We approximate the surface as a square potential barrier.



We assume the electrons outside the sample have energy $E = p^2 / 2m = \hbar^2 k^2 / 2m$

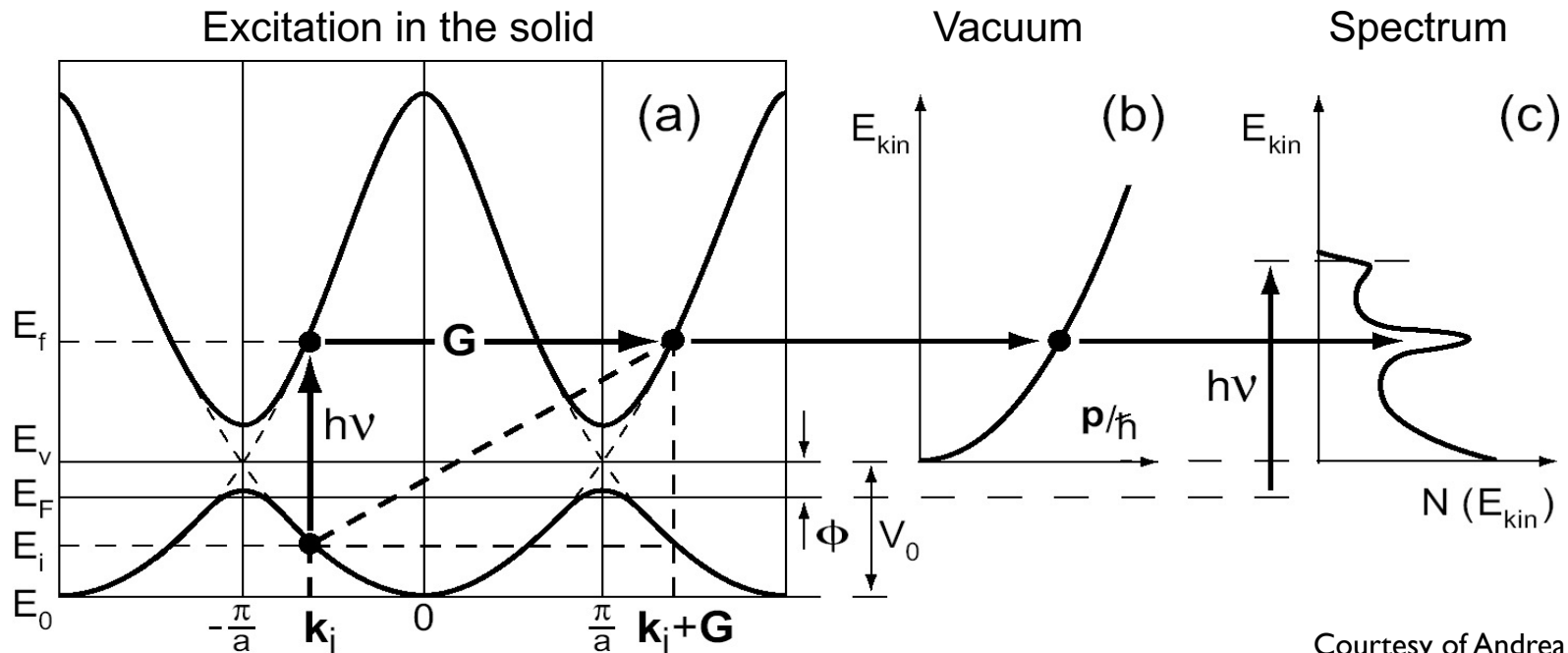


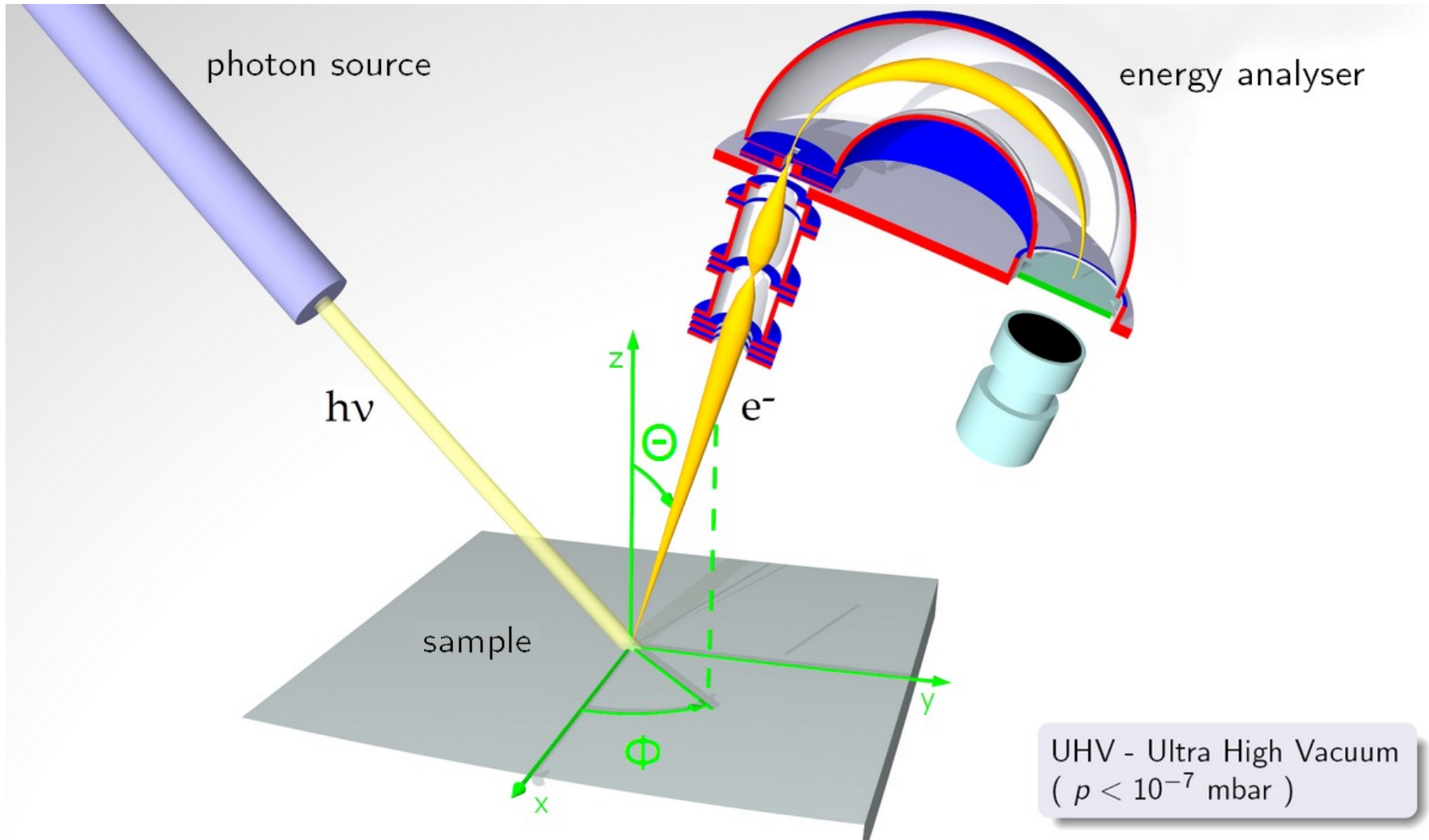
Can think of a analogous “Snell’s Law” for photoemission, where the in-plane momentum is conserved and the out-of-plane momentum changes due to scattering off the potential barrier (work function)

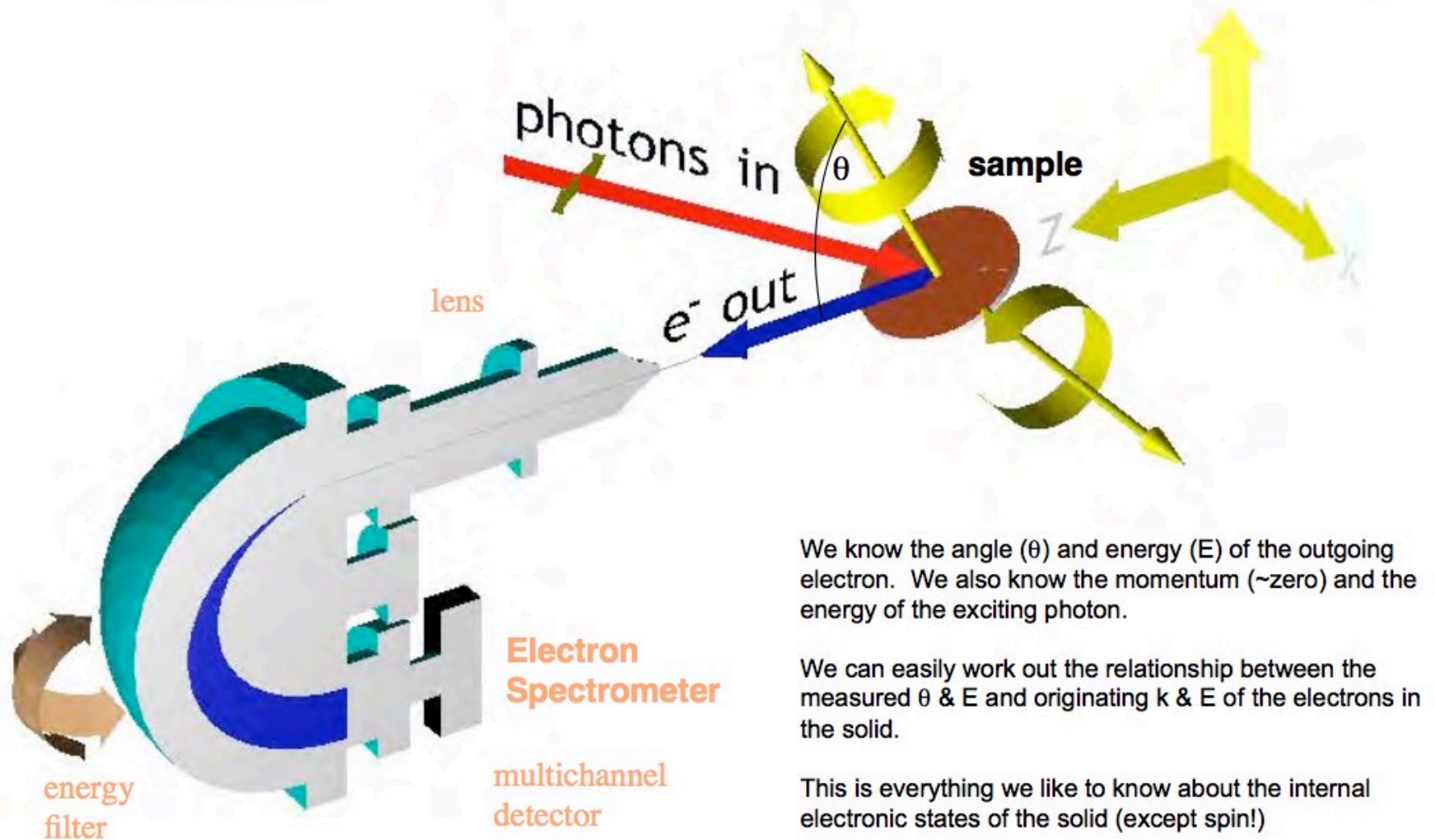
Free-electron final state $E_f(\mathbf{k}) = \frac{\hbar^2 \mathbf{k}^2}{2m} - |E_0| = \frac{\hbar^2 (\mathbf{k}_{\parallel}^2 + \mathbf{k}_{\perp}^2)}{2m} - |E_0|$

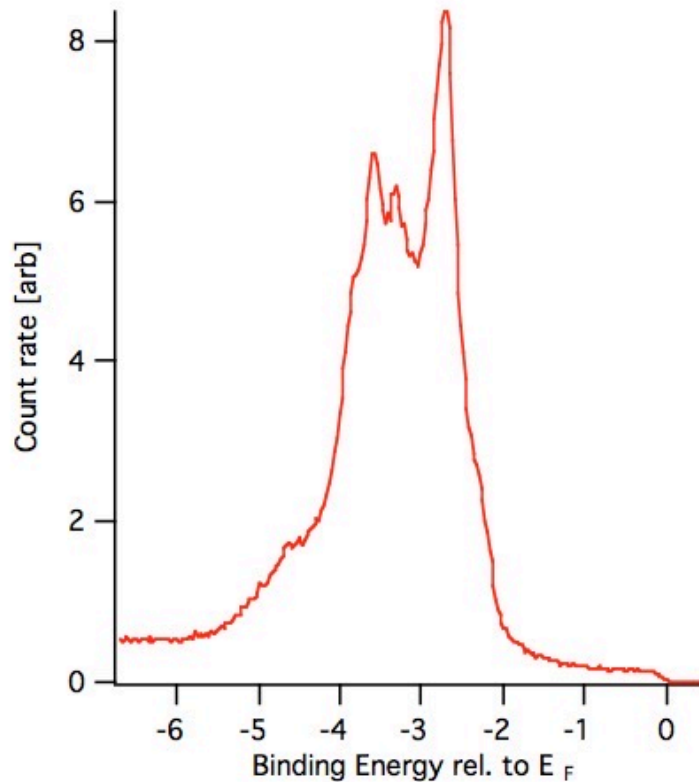
because $\frac{\hbar^2 \mathbf{k}_{\parallel}^2}{2m} = E_{kin} \sin^2 \vartheta$ $E_f = E_{kin} + \phi$ $V_0 = |E_0| + \phi$

$\mathbf{k}_{\perp} = \frac{1}{\hbar} \sqrt{2m(E_{kin} \cos^2 \vartheta + V_0)}$

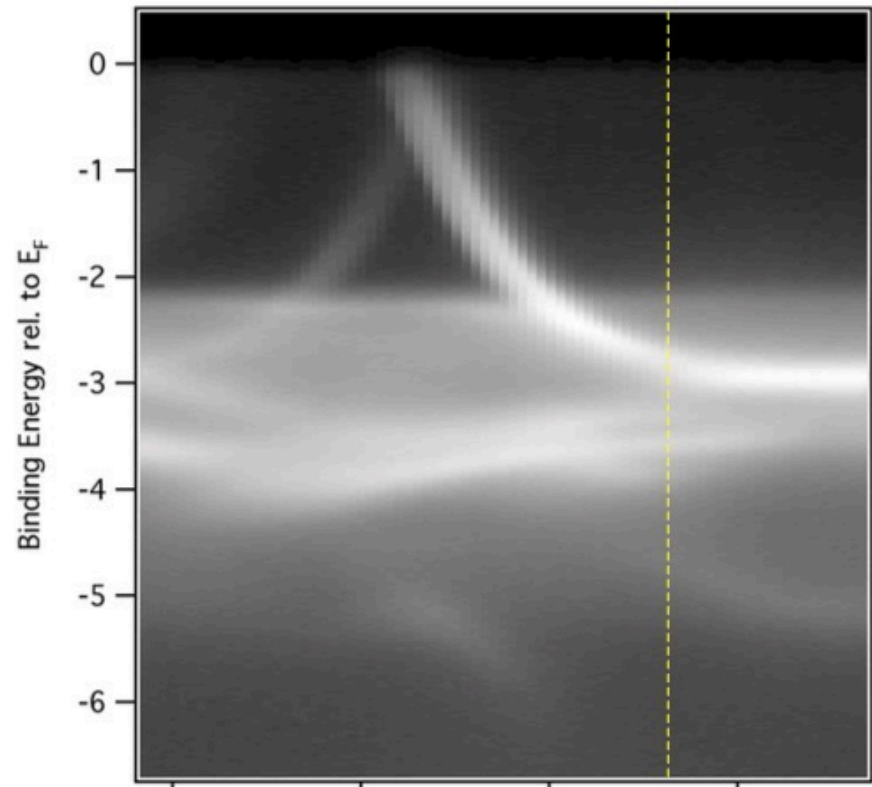






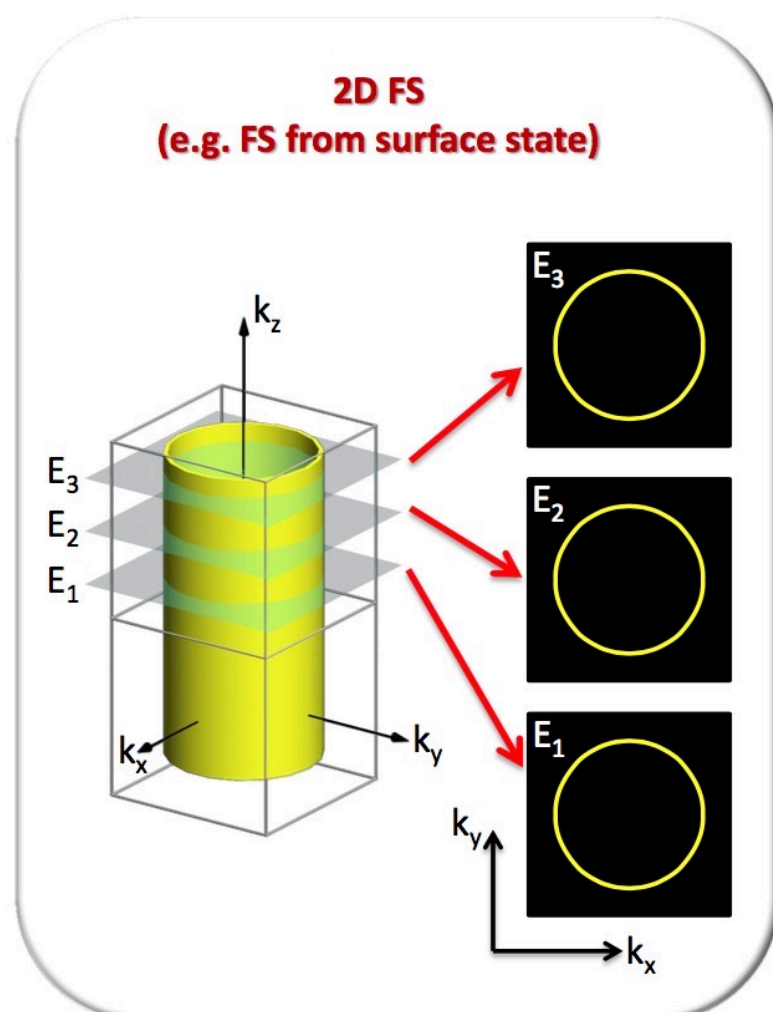
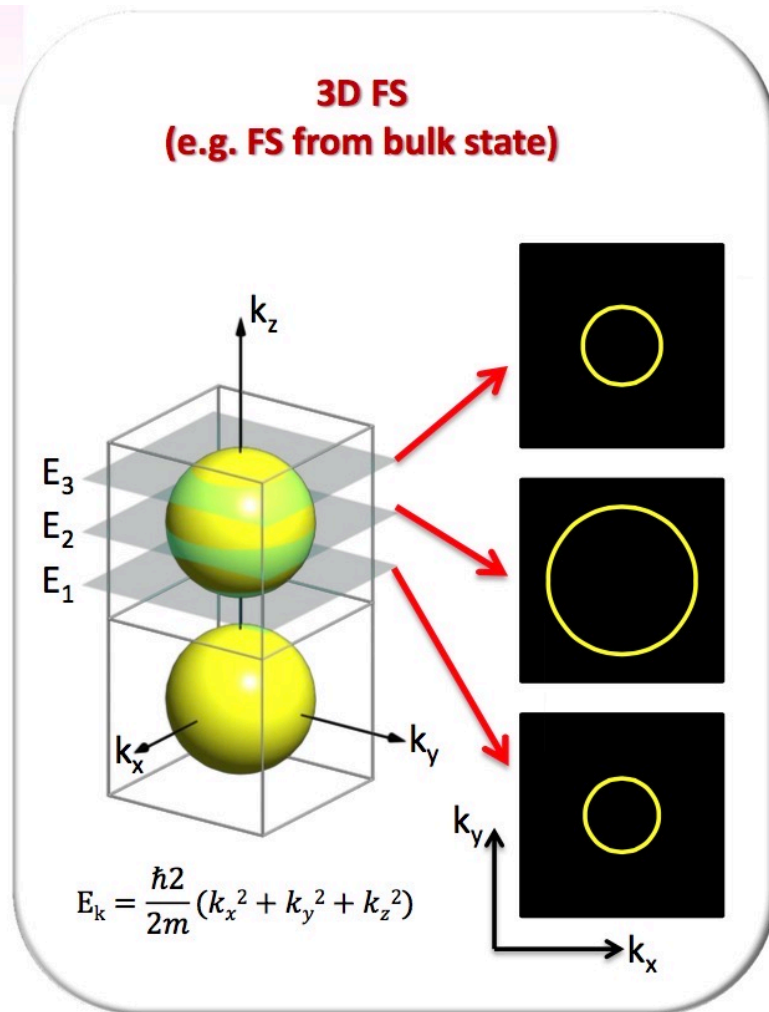


A spectrum at a single momentum k_x



Accumulate spectra as the momentum k_x is scanned

$$\mathbf{k}_{\perp} = \frac{1}{\hbar} \sqrt{2m(E_{kin} \cos^2 \vartheta + V_0)}$$



$h\nu$ (eV)	ΔE & Δk	Cross Section	Primarily Used For / Special Capabilities
10 – 100 eV synchrotrons, plasma lamps, lasers	10^{-3} eV 10^{-3} Å⁻¹	10 - 1 (Mb / atom)	High-resolution studies of electronic structure & surfaces, Fermi surface & band mapping, low-energy physics
100 – 1000 eV synchrotrons	10^{-2} eV 10^{-2} Å⁻¹	10 - 0.01 (Mb / atom)	Resonant photoemission X-ray absorption / magnetic dichroism XPS (elemental chemical analysis)
1000 – 10,000 eV X-ray tubes, synchrotrons	10^{-1} eV 10^{-1} Å⁻¹	$10^{-2} - 10^{-4}$ (Mb / atom)	Bulk sensitivity Elemental & chemical analysis Changing orbital cross-sections

X-ray tubes

Most common sources for XPS (Al, Mg anodes), can be used with grating for better energy resolution (1000-10,000 eV, $\Delta E \sim 0.1-1$ eV)

Plasma Discharge

Narrow bandwidth, high intensity lamps in VUV (10-100 eV, $\Delta E \sim 0.001$ eV); used for ARPES

Synchrotrons

Complete control over photon beam (energy, polarization, resolution); user facilities

Lasers

Higher harmonic generation; low energy (> 10 eV). Pump-probe, or high resolution

Photon source must be :

1. Monochromatic
2. High intensity ($> 10^9$ s⁻¹)
3. Energetic ($h\nu > \phi \sim 5$ eV)

X-ray photoemission measurements of core electrons

- elemental & chemical analysis of sample surfaces
- determination of oxidation states, contaminants, surface composition
- simple, user-friendly technique
- widespread use in physics, materials science, chemistry, engineering, as well as the “real world”

Photon Energy (eV)	Energy Resolution (eV)	Spatial Resolution	Found in
$10^3 - 10^4$ eV (X-Ray tubes, Synchrotrons)	$0.1 - 1$ eV	$0.01 - 1$ mm	Many universities National labs Synchrotrons Some companies

UV photoemission measurements of valence electrons

- studies of electronic band structure & density of states
- low temperature (> 10 K), high resolution (0.001 eV) measurements
- commonly studied materials : superconductors, novel electronic materials, topological materials, surfaces, nanostructures on surfaces
- a specialized technique used in solid state physics & materials science to study electronic structure & interactions (accessible at synchrotrons)

Photon Energy (eV)	Energy Resolution (eV)	Angular Resolution	Found in
10 – 100 eV (Synchrotrons, Lasers, Plasma lamps)	0.01 – 0.001 eV	0.1 – 1°	Some academic groups; Synchrotrons

Which of the following is strongly dependent on the value of the photon energy used to perform the experiments? Assume the photon energy is always large enough to eject electrons from the sample.

- | | |
|---|---------------------|
| 1. The work function of the sample | A. 2 only |
| 2. The number of (occupied) states that we can access below E_F | B. 2, 3 & 4 |
| 3. The number of (unoccupied) states that we can access above E_F | C. 1, 2, & 4 |
| | D. 2 & 4 |
| 4. The intensity of the measured photoemission signal | E. All of the above |

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Which of the following is strongly dependent on the value of the photon energy used to perform the experiments? Assume the photon energy is always large enough to eject electrons from the sample.

1. The momentum resolution of our measurement (i.e. how fine a feature Δk we can resolve)
2. The range in momentum space (span of the Brillouin zone) that we can access
3. The out-of-plane momentum, k_z , that is being accessed

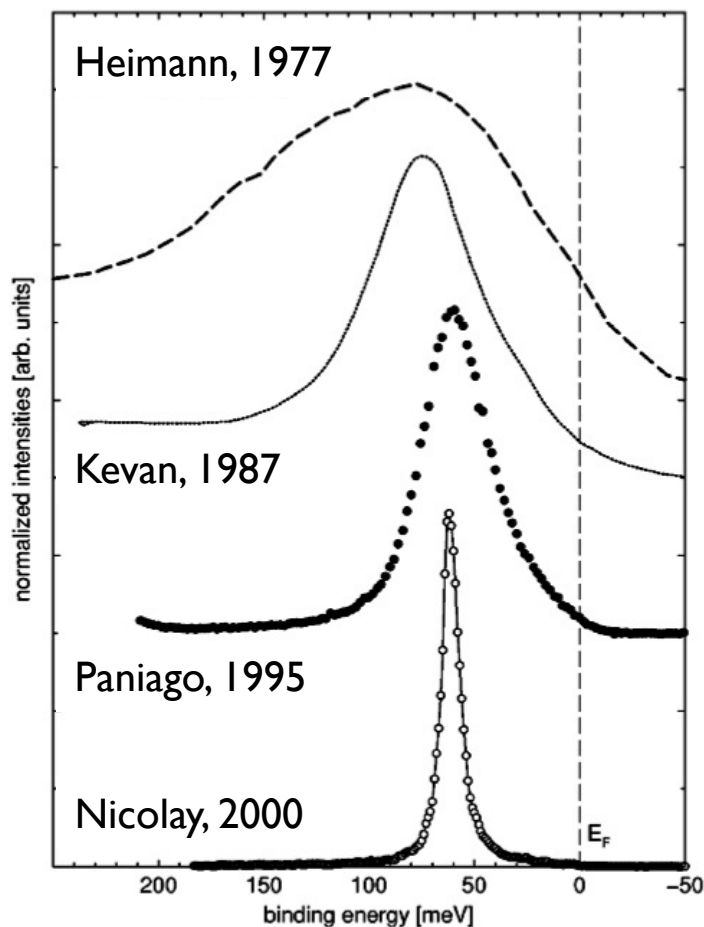
- A. None of the above
- B. 2 only
- C. 2 & 3
- D. 3 only
- E. All of the above

Which of the following is strongly dependent on the value of the photon energy used to perform the experiments? Assume the photon energy is always large enough to eject electrons from the sample.

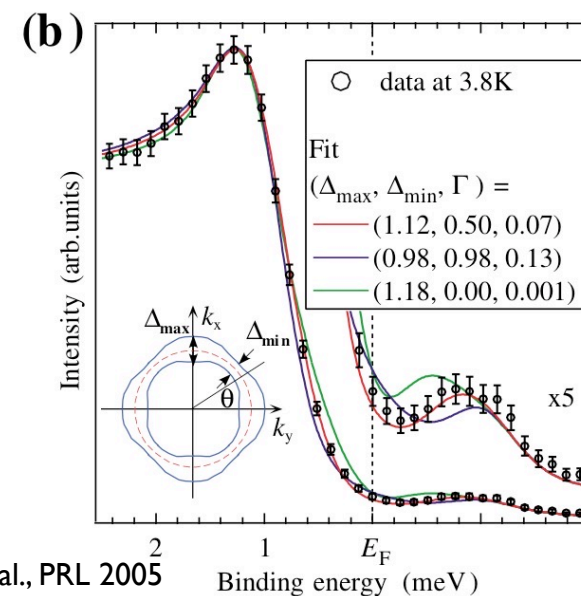
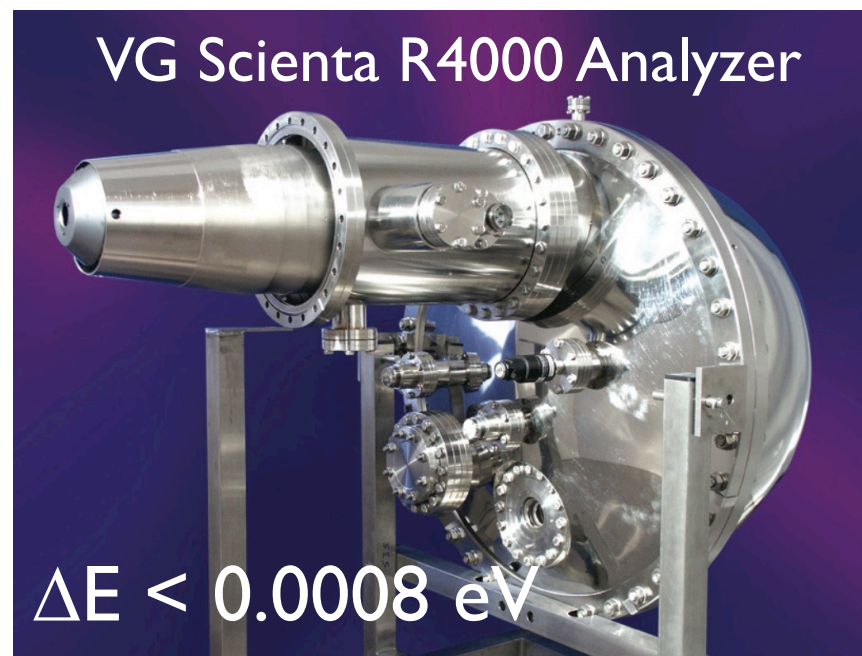
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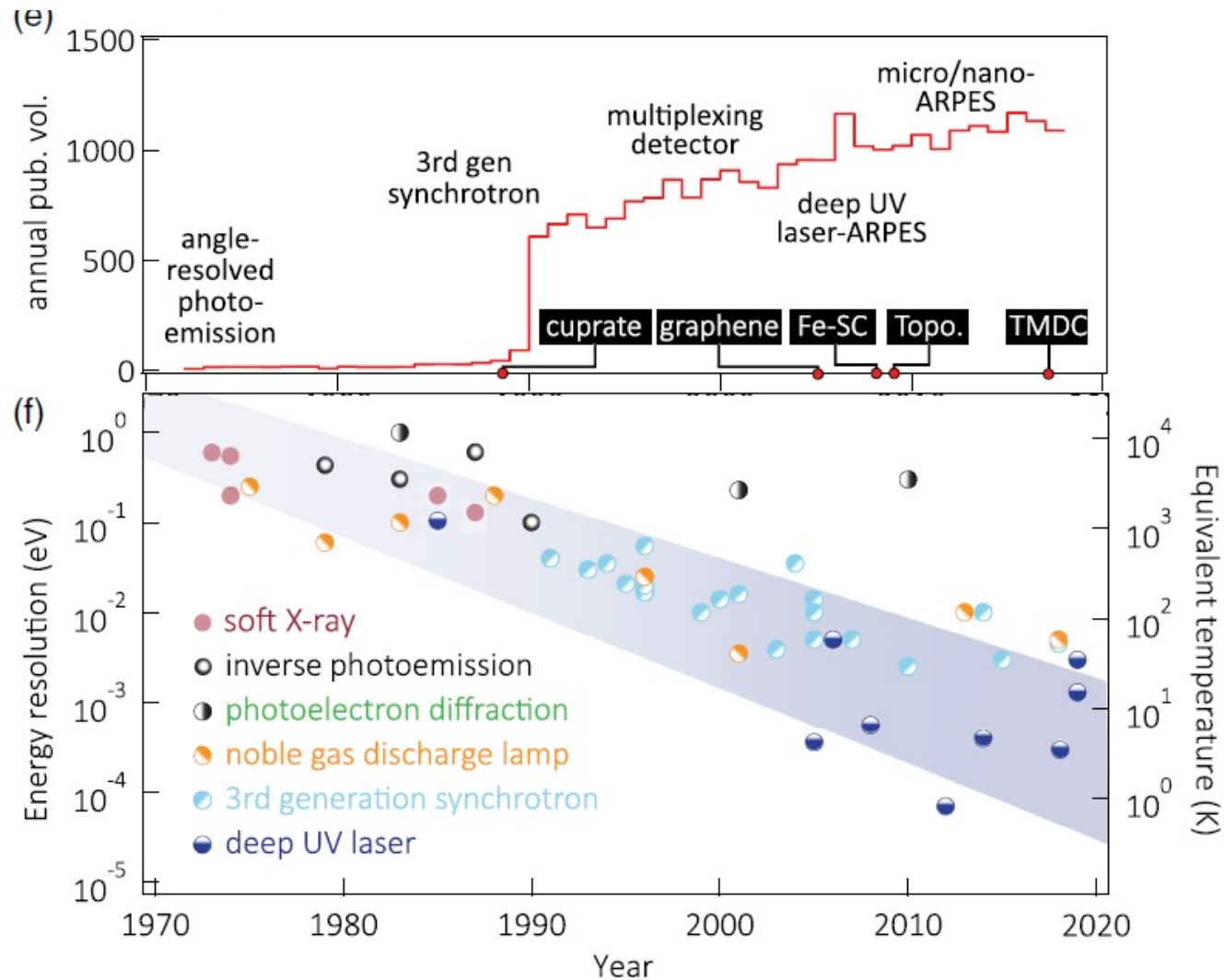
Evolution of instrumental resolution over time



F. Reinert et al., PRB (2001)



T. Kiss et al., PRL 2005



J.A. Sobota, Y. He, and Z.-X. Shen. Reviews of Modern Physics 93, 025006 (2021)

- mean free path (mfp) of photoelectrons in solids is ~ 1 **nanometer**. Measurements are sensitive to top unit cells & monolayers
- short mfp means sample surfaces must be kept pristine (no adsorbed gases) in ultrahigh vacuum (UHV - 10^{-10} torr)
- need to reduce stray electromagnetic fields (deflection)
- infeasible to measure samples *in vivo* or in solution

